

Mathematical Methods

Module 2 – Measure, Integration and L^p Spaces

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Theory, intuition, visual explanations, examples and guided exercises

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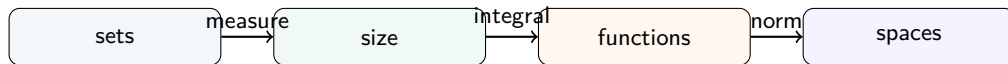
L^p spaces

Approximation, density and exam training

Module 2 in one sentence

What we are doing

We build a language where the phrase “the integral of a limit” has a precise meaning and where functions can be treated as points of a space.



- The first difficulty is not computation, but deciding what objects are admissible.
- Once admissible objects are fixed, many computations become systematic.

The central dictionary

Set language

- measurable set: admissible event or region;
- measure: size of a set;
- null set: negligible set.

Function language

- measurable function: observable quantity;
- integral: accumulated size;
- L^p norm: average magnitude.

Exercise: Quick translation

Translate the statement $f = g$ almost everywhere into ordinary words.

Answer

The two functions may differ, but only on a set whose measure is zero.

Counting before integrating

Example: Finite set

Let $X = \{a, b, c, d\}$ and let μ be counting measure. For

$$f(a) = 2, \quad f(b) = 0, \quad f(c) = 3, \quad f(d) = 1,$$

the integral is just a sum:

$$\int_X f \, d\mu = 2 + 0 + 3 + 1 = 6.$$

Lesson

Lebesgue integration generalizes ordinary summation. Continuous area is only one special case.

Integral as weighted sum

Example: Weighted counting measure

Let $X = \{1, 2, 3\}$ and let

$$\mu(\{1\}) = 0.2, \quad \mu(\{2\}) = 0.5, \quad \mu(\{3\}) = 0.3.$$

If $f(1) = 10$, $f(2) = 4$, $f(3) = 0$, then

$$\int_X f \, d\mu = 10(0.2) + 4(0.5) + 0(0.3) = 4.$$

Exercise: Interpretation

What changes if the weights do not sum to 1?

Two ways to sum a simple function

Example: Simple function

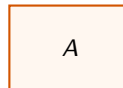
Let

$$f = 3\mathbf{1}_A + 5\mathbf{1}_B,$$

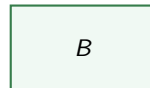
where $A \cap B = \emptyset$, $\mu(A) = 2$ and $\mu(B) = 4$. Then

$$\int f \, d\mu = 3\mu(A) + 5\mu(B) = 3 \cdot 2 + 5 \cdot 4 = 26.$$

height 3



height 5



What makes Module 2 difficult?

Three recurring sources of confusion

- ① A set can have infinitely many points and still have measure zero.
- ② A function can be changed on a null set without changing its L^p element.
- ③ Limit and integral do not always commute.

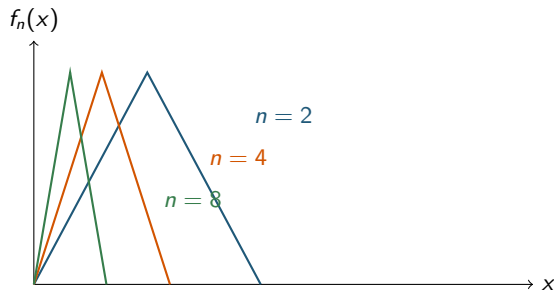
Exercise: Preview

Give an example of an infinite subset of $\mathbb{R}$ with Lebesgue measure zero.

Answer

$\mathbb{Q} \cap [0, 1]$ is countably infinite and has Lebesgue measure zero.

A visual warning: pointwise limits



Intuition

A sequence of functions may converge pointwise to zero while preserving area. This is why convergence theorems need hypotheses.

Exercise: the moving spike

Exercise: Pointwise limit versus integral

On $[0, 1]$, define

$$f_n(x) = n\mathbf{1}_{(0, 1/n)}(x).$$

Find the pointwise limit of f_n and compute $\int_0^1 f_n(x) dx$.

Solution

For every fixed $x > 0$, eventually $x > 1/n$, so $f_n(x) = 0$. At $x = 0$ the indicator is also zero. Hence $f_n \rightarrow 0$ pointwise. However

$$\int_0^1 f_n(x) dx = n\lambda((0, 1/n)) = 1.$$

Thus the integral does not converge to the integral of the pointwise limit.

Module 2

Orientation: from sets to integration

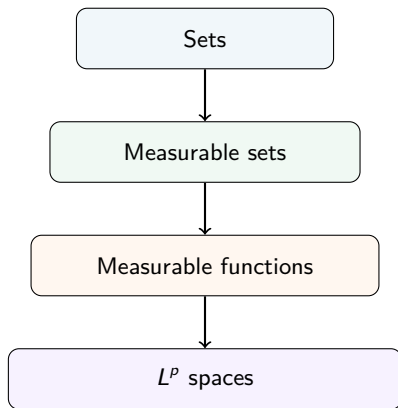
Why Module 2 is needed

The main idea

In Module 1 we studied spaces, distances, norms and inner products. Module 2 gives a rigorous way to measure the size of sets and the size of functions.

- We want to integrate functions that are not necessarily continuous.
- We want convergence theorems that justify passing limits inside integrals.
- We want function spaces where distance is measured by an integral.
- We want to understand when two functions are considered the same.

The three-level structure



- First decide which sets can be measured.
- Then define functions compatible with those sets.
- Then integrate and build normed spaces.

From Riemann to Lebesgue: intuition

Riemann viewpoint

Partition the domain into small intervals and sum rectangular areas.

- Works very well for continuous functions.
- Less flexible for irregular limits.

Lebesgue viewpoint

Measure the size of level sets and sum according to values.

- Handles discontinuities and limits better.
- Natural for spaces of functions.

A guiding example

Example: A function with many discontinuities

Let

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \cap [0, 1], \\ 0, & x \in [0, 1] \setminus \mathbb{Q}. \end{cases}$$

This function is discontinuous at every point of $[0, 1]$.

- Riemann integral: does not exist.
- Lebesgue integral: exists and equals 0, since $\mathbb{Q} \cap [0, 1]$ has length 0.
- Moral: the size of the set where the function is nonzero matters.

What students should be able to do

- ① Recognize sigma-algebras and measurable sets.
- ② Compute measures of basic sets and use continuity of measures.
- ③ Prove measurability of functions through preimages and operations.
- ④ Compute Lebesgue integrals in concrete examples.
- ⑤ Apply Monotone Convergence, Fatou and Dominated Convergence correctly.
- ⑥ Work with L^p norms and inequalities.
- ⑦ Produce counterexamples when hypotheses are missing.
- ⑧ Solve short proof-based and calculation-based exam questions.

Notation for Module 2

Sets and measures

- X : abstract set.
- $\mathcal{F}$: sigma-algebra.
- $(X, \mathcal{F})$: measurable space.
- μ : measure.
- $(X, \mathcal{F}, \mu)$: measure space.

Functions and spaces

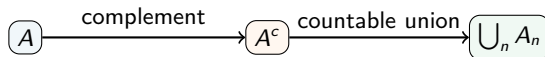
- $\mathbf{1}_A$: indicator of A .
- $f_n \rightarrow f$ a.e.: almost everywhere.
- $L^p(X, \mu)$: integrable power space.
- $\|f\|_p$: L^p norm.

Sigma-algebras: intuition before axioms

Why the axioms look the way they do

If a set is admissible, we also want to admit:

- its complement: “not in the set”;
- countable unions: limits of repeated operations;
- countable intersections: obtained from unions by complements.



Finite example with atoms

Example: Partition as information

Let $X = \{1, 2, 3, 4, 5, 6\}$ and suppose we only distinguish the blocks

$$P_1 = \{1, 2\}, \quad P_2 = \{3, 4\}, \quad P_3 = \{5, 6\}.$$

Then every measurable set is a union of whole blocks.



Exercise: atoms determine the sigma-algebra

Exercise: Generated by two sets

Let $X = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$ and $B = \{3, 4\}$. Find the atoms of $\sigma(A, B)$.

Solution

Separate points by their membership in A and B :

$$A \cap B = \{3\}, \quad A \cap B^c = \{1, 2\},$$

$$A^c \cap B = \{4\}, \quad A^c \cap B^c = \{5\}.$$

Hence the atoms are $\{3\}, \{1, 2\}, \{4\}, \{5\}$ and the generated sigma-algebra has $2^4 = 16$ sets.

Checklist: is it a sigma-algebra?

Operational checklist

To check whether $\mathcal{F}$ is a sigma-algebra on X , verify:

- ① Is $X \in \mathcal{F}$?
- ② If $A \in \mathcal{F}$, is $A^c \in \mathcal{F}$?
- ③ If $A_1, A_2, \dots \in \mathcal{F}$, is $\bigcup_n A_n \in \mathcal{F}$?

Counterexample: Common shortcut that is not enough

Checking only finite unions gives an algebra, not necessarily a sigma-algebra.

Exercise: find the broken axiom

Exercise: Non-example

Let $X = \{1, 2, 3, 4\}$ and

$$\mathcal{C} = \{\emptyset, X, \{1\}, \{2, 3, 4\}, \{2\}\}.$$

Is $\mathcal{C}$ a sigma-algebra?

Solution

No. Since $\{2\} \in \mathcal{C}$, its complement $\{1, 3, 4\}$ should belong to $\mathcal{C}$, but it does not. The complement axiom fails.

Exercise: an algebra that is not a sigma-algebra

Exercise: Finite-cofinite sets

Let $X = \mathbb{N}$ and let $\mathcal{A}$ be the class of finite or cofinite subsets of $\mathbb{N}$. Show that $\mathcal{A}$ is not a sigma-algebra.

Solution

Each singleton $\{2n\}$ is finite, hence belongs to $\mathcal{A}$. But

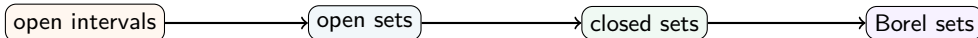
$$\bigcup_{n \geq 1} \{2n\} = \{2, 4, 6, \dots\}$$

is infinite and its complement is infinite. Therefore it is neither finite nor cofinite, so it is not in $\mathcal{A}$.

Borel sets: how to build many sets

From intervals to complicated sets

Starting from open intervals and applying countable unions, intersections and complements, we obtain Borel sets.



- Open sets in $\mathbb{R}$ are countable unions of open intervals.
- Closed sets are complements of open sets.

Exercise: write a set using Borel operations

Exercise: Singletons are Borel

Show that $\{a\}$ is a Borel subset of $\mathbb{R}$.

Solution

One representation is

$$\{a\} = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, a + \frac{1}{n} \right).$$

Each interval is open, hence Borel. A countable intersection of Borel sets is Borel.

Exercise: countable sets are Borel

Exercise: Generalize the singleton argument

Let $C = \{c_1, c_2, c_3, \dots\} \subset \mathbb{R}$ be countable. Prove that C is Borel.

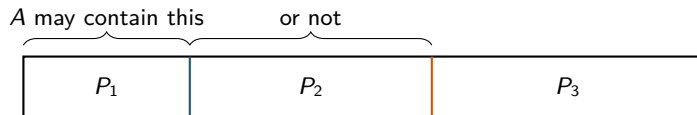
Solution

Since each singleton $\{c_n\}$ is Borel,

$$C = \bigcup_{n=1}^{\infty} \{c_n\}$$

is a countable union of Borel sets. Hence $C \in \mathcal{B}(\mathbb{R})$.

Visual: generating a sigma-algebra by a partition



measurable sets are unions of atoms

Key point

A sigma-algebra represents the distinctions that are observable. Atoms are the finest distinguishable pieces.

Common mistake: “all subsets are Borel

Counterexample: False statement

It is false that every subset of $\mathbb{R}$ is Borel.

What students need to know here

For this course it is enough to remember:

- all intervals are Borel;
- all open and closed sets are Borel;
- all countable sets are Borel;
- complements and countable unions/intersections preserve Borelness.

Exam-style: prove closure under intersections

Exam-style exercise: Proof task

Let $\mathcal{F}$ be a sigma-algebra. Prove that if $A_n \in \mathcal{F}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$.

Solution

By De Morgan's law,

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c.$$

Since $A_n \in \mathcal{F}$, also $A_n^c \in \mathcal{F}$. The countable union is in $\mathcal{F}$, and taking the complement keeps us in $\mathcal{F}$.

Module 2

Sigma-algebras and Borel sets

Why not measure every subset?

The problem

On $\mathbb{R}$ there is no measure defined on all subsets that is simultaneously:

- translation-invariant,
 - countably additive,
 - equal to interval length on intervals.
-
- We therefore choose a class of admissible sets.
 - This class must be stable under the operations used in limits.
 - The correct structure is a sigma-algebra.

Definition: Algebra

A class $\mathcal{A} \subseteq \mathcal{P}(X)$ is an algebra if:

- ① $X \in \mathcal{A}$;
- ② $A \in \mathcal{A} \Rightarrow A^c \in \mathcal{A}$;
- ③ $A, B \in \mathcal{A} \Rightarrow A \cup B \in \mathcal{A}$.

- An algebra is automatically closed under finite intersections and finite differences.
- It is not necessarily closed under countable unions.

Definition: Sigma-algebra

A class $\mathcal{F} \subseteq \mathcal{P}(X)$ is a sigma-algebra if:

- ① $X \in \mathcal{F}$;
- ② $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$;
- ③ if $A_n \in \mathcal{F}$ for all n , then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

- The pair $(X, \mathcal{F})$ is called a measurable space.
- Elements of $\mathcal{F}$ are called measurable sets.

Immediate consequences

Theorem: Closure properties

If $\mathcal{F}$ is a sigma-algebra, then:

$$\emptyset \in \mathcal{F}, \quad \bigcap_{n=1}^{\infty} A_n \in \mathcal{F}, \quad A \setminus B \in \mathcal{F}.$$

Proof idea: Proof idea

Use complements and De Morgan's laws:

$$\bigcap_n A_n = \left(\bigcup_n A_n^c \right)^c.$$

Also $\emptyset = X^c$ and $A \setminus B = A \cap B^c$.

Smallest and largest sigma-algebras

Example: Two extremes

For any set X :

$$\{\emptyset, X\}$$

is the smallest sigma-algebra, and

$$\mathcal{P}(X)$$

is the largest sigma-algebra.

- The smallest one contains almost no information.
- The largest one contains every subset.
- In applications we usually work between these two extremes.

Example: a finite sigma-algebra

Example: Partition-generated sigma-algebra

Let $X = \{1, 2, 3, 4, 5, 6\}$ and partition

$$P_1 = \{1, 2\}, \quad P_2 = \{3\}, \quad P_3 = \{4, 5, 6\}.$$

The sigma-algebra generated by the partition contains all unions of the atoms P_1, P_2, P_3 .

- Number of measurable sets: $2^3 = 8$.
- Each measurable set is made by selecting atoms.

Exercise: list the sigma-algebra

Exercise: Partition atoms

With the partition

$$P_1 = \{1, 2\}, \quad P_2 = \{3\}, \quad P_3 = \{4, 5, 6\},$$

write all sets in the generated sigma-algebra.

Answer

$$\emptyset, P_1, P_2, P_3, P_1 \cup P_2, P_1 \cup P_3, P_2 \cup P_3, X.$$

Generated sigma-algebra

Definition: Sigma-algebra generated by a class

Let $\mathcal{C} \subseteq \mathcal{P}(X)$. The sigma-algebra generated by $\mathcal{C}$ is

$$\sigma(\mathcal{C}) = \bigcap \{ \mathcal{F} : \mathcal{F} \text{ is a sigma-algebra and } \mathcal{C} \subseteq \mathcal{F} \}.$$

- It is the smallest sigma-algebra containing $\mathcal{C}$.
- It contains exactly the sets forced by countable set operations.

Proof: generated sigma-algebra exists

Proof idea: Intersection argument

- ① At least one sigma-algebra contains $\mathcal{C}$: namely $\mathcal{P}(X)$.
- ② The intersection of any family of sigma-algebras is again a sigma-algebra.
- ③ Therefore the intersection of all sigma-algebras containing $\mathcal{C}$ is the smallest such sigma-algebra.

Exercise: Check the key step

Prove directly that an arbitrary intersection of sigma-algebras is a sigma-algebra.

Borel sigma-algebra on the real line

Definition: Borel sets

The Borel sigma-algebra on $\mathbb{R}$ is

$$\mathcal{B}(\mathbb{R}) = \sigma(\{(a, b) : a < b\}).$$

Elements of $\mathcal{B}(\mathbb{R})$ are called Borel sets.

- Open intervals are Borel.
- Open sets, closed sets, half-lines and countable sets are Borel.
- Borel sets are rich enough for most analytical purposes.

Equivalent generators of $\mathcal{B}(\mathbb{R})$

Theorem: Many generating classes

The following classes generate the same Borel sigma-algebra:

$$\begin{aligned} &\{(a, b) : a < b\}, \quad \{(-\infty, a) : a \in \mathbb{R}\}, \\ &\{(-\infty, a] : a \in \mathbb{R}\}, \quad \{(a, \infty) : a \in \mathbb{R}\}. \end{aligned}$$

Proof idea: Typical step

For example,

$$(a, b) = (-\infty, b) \cap (-\infty, a]^c.$$

So intervals are generated by half-lines, and conversely half-lines are countable unions of intervals.

Exercise: Borel or not?

Exercise: Classify

Decide whether the following sets are Borel:

- ① $\mathbb{N}$.
- ② $\mathbb{Q}$.
- ③ $\mathbb{R} \setminus \mathbb{Q}$.
- ④ $\bigcap_{n \geq 1} (-1/n, 1/n)$.
- ⑤ $\bigcup_{n \geq 1} [n, n + 1/n]$.

Answer

All are Borel. Use countability and closure under countable unions and intersections.

Counterexample: countable unions matter

Counterexample: Algebra but not sigma-algebra

Let $X = \mathbb{N}$ and let $\mathcal{A}$ be the class of finite or cofinite subsets of $\mathbb{N}$. Then $\mathcal{A}$ is an algebra but not a sigma-algebra.

- Singletons $\{2n\}$ are finite, hence in $\mathcal{A}$.
- Their countable union is the set of even integers.
- This set is neither finite nor cofinite.

Monotone classes: useful idea

Definition: Monotone class

A class $\mathcal{M}$ is a monotone class if it is closed under:

- increasing limits $A_n \uparrow A$;
- decreasing limits $A_n \downarrow A$.

Why useful?

Many results are proved first on simple sets, then extended by a monotone-class argument.

Exam-style exercise: generated sigma-algebra

Exam-style exercise: Finite generated sigma-algebra

Let $X = \{1, 2, 3, 4\}$ and let

$$A = \{1, 2\}, \quad B = \{2, 3\}.$$

Find $\sigma(A, B)$.

Solution idea

Find the atoms determined by membership in A and B :

$$\{1\}, \quad \{2\}, \quad \{3\}, \quad \{4\}.$$

Hence $\sigma(A, B) = \mathcal{P}(X)$.

Exam-style exercise: non-example

Exam-style exercise: Check the axioms

Let $X = \{1, 2, 3\}$ and

$$\mathcal{C} = \{\emptyset, \{1\}, \{2, 3\}, X, \{2\}\}.$$

Is $\mathcal{C}$ a sigma-algebra?

Solution

No. Since $\{2\} \in \mathcal{C}$, its complement $\{1, 3\}$ should belong to $\mathcal{C}$, but it does not.

Measure as a consistent notion of size

What a measure must do

A measure assigns a size to admissible sets and behaves additively on disjoint pieces.



$$A \cap B = \emptyset$$

$$\mu(A \cup B) = \mu(A) + \mu(B)$$

Counting measure: computations

Example: Counting measure on a finite set

Let $X = \{1, 2, 3, 4, 5\}$ and $A = \{1, 4, 5\}$. Then

$$\mu(A) = 3.$$

If $B = \{2, 5\}$, then

$$\mu(A \cup B) = \mu(\{1, 2, 4, 5\}) = 4.$$

Exercise: Check inclusion-exclusion

Verify that $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$.

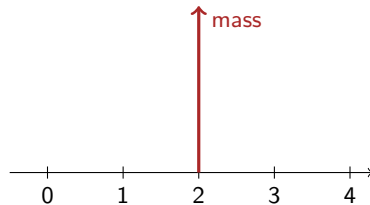
Dirac measure: all mass at one point

Example: Dirac measure

Fix $x_0 = 2$ in $\mathbb{R}$. Then

$$\delta_2(A) = \begin{cases} 1, & 2 \in A, \\ 0, & 2 \notin A. \end{cases}$$

Thus $\delta_2((1, 3)) = 1$ and $\delta_2((3, 4)) = 0$.



Weighted counting measure

Example: Discrete but not uniform

On $\mathbb{N}$, define

$$\mu(A) = \sum_{n \in A} 2^{-n}.$$

Then

$$\mu(\{1, 3, 5\}) = 2^{-1} + 2^{-3} + 2^{-5} = \frac{1}{2} + \frac{1}{8} + \frac{1}{32}.$$

Practical meaning

Some points may count more than others. The integral will become a weighted sum.

Finite versus sigma-finite: intuition

Finite

The whole space has finite size:

$$\mu(X) < \infty.$$

Example: $[0, 1]$ with length.

Sigma-finite

The space may be infinite, but can be covered by countably many finite-size pieces:

$$X = \bigcup_n X_n, \quad \mu(X_n) < \infty.$$

Example: $\mathbb{R} = \bigcup_n [-n, n]$.

Exercise: sigma-finite or not?

Exercise: Classify

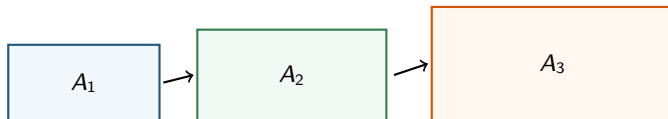
Decide whether each measure space is finite, sigma-finite, or neither.

- ① $([0, 1], \mathcal{B}([0, 1]), \lambda)$.
- ② $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$.
- ③ An uncountable set X with counting measure on $\mathcal{P}(X)$.

Solution

(1) finite. (2) not finite but sigma-finite. (3) not sigma-finite: any finite-measure set is finite, and a countable union of finite sets is countable, not all of X .

Continuity from below: visual idea



$$A_1 \subset A_2 \subset A_3 \subset \dots$$

Theorem: Continuity from below

If $A_n \uparrow A$, then $\mu(A_n) \uparrow \mu(A)$.

Continuity from above: why a hypothesis is needed

Counterexample: Infinite starting measure causes trouble

Let $A_n = (n, \infty)$ in $\mathbb{R}$. Then

$$A_n \downarrow \emptyset,$$

but

$$\lambda(A_n) = \infty$$

for every n . Thus $\lambda(A_n)$ does not decrease to 0.

Lesson

For decreasing sets we need $\mu(A_1) < \infty$.

Exercise: compute by continuity from below

Exercise: Increasing intervals

Let $A_n = (0, 1 - 1/n)$ for $n \geq 2$. Find $\lim_n \lambda(A_n)$.

Solution

The sets increase to $(0, 1)$:

$$A_n \uparrow (0, 1).$$

Therefore

$$\lim_{n \rightarrow \infty} \lambda(A_n) = \lambda((0, 1)) = 1.$$

Exercise: compute by continuity from above

Exercise: Decreasing intervals

Let $A_n = [0, 1/n]$. Find $\lim_n \lambda(A_n)$.

Solution

We have $A_n \downarrow \{0\}$ and $\lambda(A_1) = 1 < \infty$. Hence

$$\lambda(A_n) \downarrow \lambda(\{0\}) = 0.$$

Indeed $\lambda([0, 1/n]) = 1/n \rightarrow 0$.

Inclusion-exclusion: a concrete example

Example: Two intervals

Let $A = [0, 2]$ and $B = [1, 4]$. Then

$$A \cup B = [0, 4], \quad A \cap B = [1, 2].$$

Thus

$$\lambda(A \cup B) = 4,$$

and

$$\lambda(A) + \lambda(B) - \lambda(A \cap B) = 2 + 3 - 1 = 4.$$

Null sets: big in points, small in measure

Main idea

A set can contain infinitely many points and still have measure zero.

Example: Countable sets

Every singleton has Lebesgue measure zero. Therefore every countable set has measure zero:

$$\lambda\left(\bigcup_{n=1}^{\infty}\{x_n\}\right) \leq \sum_{n=1}^{\infty} \lambda(\{x_n\}) = 0.$$

Exercise: Consequence

What is $\lambda(\mathbb{Q} \cap [0, 1])$?

Exam-style: prove subadditivity

Exam-style exercise: Proof task

Prove that for measurable sets A_n ,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

Solution

Define disjoint sets

$$B_1 = A_1, \quad B_n = A_n \setminus \bigcup_{k < n} A_k.$$

Then $B_n \subset A_n$, the B_n are disjoint, and $\bigcup_n B_n = \bigcup_n A_n$. Hence

$$\mu\left(\bigcup_n A_n\right) = \sum_n \mu(B_n) \leq \sum_n \mu(A_n).$$

Module 2

Measures and measure spaces

Measures: the abstract notion of size

Definition: Measure

Let $(X, \mathcal{F})$ be a measurable space. A measure is a map

$$\mu : \mathcal{F} \rightarrow [0, \infty]$$

such that:

- ① $\mu(\emptyset) = 0$;
- ② if (A_n) are disjoint measurable sets, then

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mu(A_n).$$

- The second property is countable additivity.

Examples of measures

Example: Counting measure

On any set X :

$$\mu(A) = \#A,$$

with value $+\infty$ if A is infinite.

Example: Dirac measure

Fix $x_0 \in X$:

$$\delta_{x_0}(A) = \begin{cases} 1, & x_0 \in A, \\ 0, & x_0 \notin A. \end{cases}$$

Example: Lebesgue measure

On $\mathbb{R}$, $\lambda((a, b)) = b - a$.

Example: Weighted counting measure

On $\mathbb{N}$:

$$\mu(A) = \sum_{n \in A} w_n, \quad w_n \geq 0.$$

Finite, sigma-finite and probability measures

Definition: Size classes

Let $(X, \mathcal{F}, \mu)$ be a measure space.

- μ is finite if $\mu(X) < \infty$.
- μ is sigma-finite if $X = \bigcup_n X_n$ with $\mu(X_n) < \infty$.
- μ is a probability measure if $\mu(X) = 1$.

Example: Lebesgue measure

Lebesgue measure on $\mathbb{R}$ is not finite, but it is sigma-finite because

$$\mathbb{R} = \bigcup_{n=1}^{\infty} [-n, n].$$

Basic properties of measures

Theorem: Consequences of countable additivity

If $A, B \in \mathcal{F}$, then:

- ① Monotonicity: $A \subseteq B \Rightarrow \mu(A) \leq \mu(B)$.
- ② Finite additivity: if $A \cap B = \emptyset$, then $\mu(A \cup B) = \mu(A) + \mu(B)$.
- ③ Subadditivity:

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

Proof: monotonicity

Proof idea: Monotonicity

Suppose $A \subseteq B$. Then

$$B = A \cup (B \setminus A),$$

and the union is disjoint. Hence

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A).$$

Exercise: Use the same idea

Prove finite additivity for three pairwise disjoint sets.

Inclusion-exclusion for two sets

Theorem: Finite formula

If $\mu(A), \mu(B) < \infty$, then

$$\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B).$$

Proof idea: Decompose into disjoint pieces

Use

$$A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B).$$

Continuity from below

Theorem: Increasing sets

If $A_1 \subseteq A_2 \subseteq \dots$ and $A = \bigcup_n A_n$, then

$$\mu(A_n) \uparrow \mu(A).$$

Proof idea: Disjoint decomposition

Let $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$. Then the B_n are disjoint and

$$A = \bigcup_{n=1}^{\infty} B_n.$$

Continuity from above

Theorem: Decreasing sets

If $A_1 \supseteq A_2 \supseteq \cdots$, $A = \bigcap_n A_n$, and $\mu(A_1) < \infty$, then

$$\mu(A_n) \downarrow \mu(A).$$

Counterexample: Why the finite assumption matters

Let $A_n = (n, \infty)$ in $\mathbb{R}$ with Lebesgue measure. Then $A_n \downarrow \emptyset$, but

$$\lambda(A_n) = \infty \not\downarrow 0.$$

Exercise: continuity of measure

Exercise: Compute limits of measures

With Lebesgue measure on $\mathbb{R}$, compute:

- ① $\lim_n \lambda([0, 1 - 1/n])$.
- ② $\lim_n \lambda((0, 1/n))$.
- ③ $\lim_n \lambda((-n, n))$.

Answers

1, 0, $+\infty$.

Null sets

Definition: Null set

A measurable set N is null if

$$\mu(N) = 0.$$

- Countable unions of null sets are null.
- A property holds almost everywhere if it fails only on a null set.
- We write “a.e.” for almost everywhere.

Example: Singletons in $\mathbb{R}$

$$\lambda(\{x\}) = 0.$$

Hence every countable subset of $\mathbb{R}$ has Lebesgue measure zero.

Almost everywhere equality

Definition: Equality almost everywhere

Two measurable functions f, g are equal almost everywhere if

$$\mu(\{x : f(x) \neq g(x)\}) = 0.$$

Example: A function equal to zero a.e.

$$f(x) = \mathbf{1}_{\mathbb{Q} \cap [0,1]}(x)$$

is equal to zero a.e. on $[0, 1]$ with Lebesgue measure.

Completeness of a measure space

Definition: Complete measure space

A measure space is complete if every subset of every null measurable set is measurable.

- Lebesgue measure is complete.
- Borel measure on $\mathcal{B}(\mathbb{R})$ is not complete.
- Completion adds all subsets of null sets.

Counterexample: Subsets of null Borel sets

There exist subsets of the Cantor set that are not Borel. Since the Cantor set has measure zero, Borel measure is not complete.

Lebesgue measure: essential facts

Facts to remember

Lebesgue measure λ on $\mathbb{R}$ satisfies:

- $\lambda((a, b)) = b - a$.
- Translation invariance: $\lambda(A + c) = \lambda(A)$.
- Countable additivity on measurable sets.
- Countable sets have measure zero.
- Open sets are countable unions of disjoint open intervals.

Exercise: measure of common sets

Exercise: Intervals and countable sets

Compute the Lebesgue measure of:

- ① $[0, 1] \cup [2, 4]$.
- ② $\mathbb{Q} \cap [0, 1]$.
- ③ $[0, 1] \setminus \mathbb{Q}$.
- ④ $\{1/n : n \in \mathbb{N}\}$.

Answers

3, 0, 1, 0.

Counterexample: finite additivity is not enough

Counterexample: A bad set function

On $\mathbb{N}$, define

$$\nu(A) = \begin{cases} 0, & A \text{ finite,} \\ 1, & A \text{ infinite.} \end{cases}$$

This is not a measure.

- Each singleton has value 0.
- But

$$\nu(\mathbb{N}) = 1 \neq \sum_{n=1}^{\infty} \nu(\{n\}) = 0.$$

- Countable additivity fails.

Exam-style exercise: continuity from above

Exam-style exercise: Decreasing sequence

Let $(X, \mathcal{F}, \mu)$ be a measure space and let $A_n \downarrow A$ with $\mu(A_1) < \infty$. Prove that $\mu(A_n) \rightarrow \mu(A)$.

Solution skeleton

Set $B_n = A_1 \setminus A_n$. Then $B_n \uparrow A_1 \setminus A$. Use continuity from below:

$$\mu(B_n) \uparrow \mu(A_1 \setminus A).$$

Since $\mu(A_n) = \mu(A_1) - \mu(B_n)$, the claim follows.

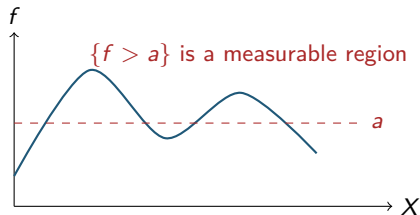
Measurability: intuition

A function is measurable if it respects observable sets

A function $f : X \rightarrow \mathbb{R}$ is measurable when sets of the form

$$\{x : f(x) > a\}, \quad \{x : f(x) \leq a\}$$

are measurable for every threshold a .



Checking measurability with thresholds

Example: Indicator functions

Let $A \in \mathcal{F}$ and define $f = \mathbf{1}_A$. For any $a \in \mathbb{R}$,

$$\{f > a\} = \begin{cases} X, & a < 0, \\ A, & 0 \leq a < 1, \\ \emptyset, & a \geq 1. \end{cases}$$

All these sets are measurable, so $\mathbf{1}_A$ is measurable.

Exercise: indicator is measurable iff the set is measurable

Exercise: Two directions

Show that $\mathbf{1}_A$ is measurable if and only if A is measurable.

Solution

If $A \in \mathcal{F}$, the previous slide proves measurability. Conversely, if $\mathbf{1}_A$ is measurable, then

$$A = \{x : \mathbf{1}_A(x) > 1/2\}$$

is measurable. Hence $A \in \mathcal{F}$.

Simple functions: stepwise measurable functions

Definition: Simple function

A simple function has the form

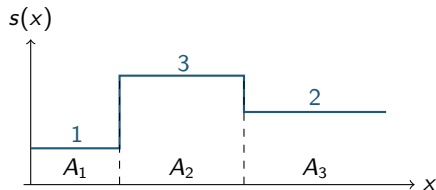
$$s(x) = \sum_{k=1}^m a_k \mathbf{1}_{A_k}(x),$$

where $a_k \in \mathbb{R}$ and $A_k \in \mathcal{F}$.

Intuition

A simple function is a finite staircase. Lebesgue integration starts from staircases and then takes limits.

Visual: a simple function



Example: Integral

If $\lambda(A_1) = .25$, $\lambda(A_2) = .35$, $\lambda(A_3) = .40$, then

$$\int s \, d\lambda = 1(.25) + 3(.35) + 2(.40) = 2.10.$$

Operations preserve measurability

Theorem: Useful closure rules

If f, g are measurable and $a, b \in \mathbb{R}$, then

$$af + bg, \quad fg, \quad |f|, \quad \max(f, g), \quad \min(f, g)$$

are measurable whenever the expressions are finite.

Practical consequence

Once basic building blocks are measurable, many complicated formulas are measurable automatically.

Exercise: measurability by composition

Exercise: Continuous transformation

Let $f : X \rightarrow \mathbb{R}$ be measurable. Prove that $g(x) = e^{f(x)}$ is measurable.

Solution

The exponential map $\varphi(t) = e^t$ is continuous, hence Borel measurable. The composition of a measurable function with a Borel measurable function is measurable. Therefore $g = \varphi \circ f$ is measurable.

Almost everywhere equality

Definition: Almost everywhere

We say $f = g$ almost everywhere, written $f = g$ a.e., if

$$\mu(\{x : f(x) \neq g(x)\}) = 0.$$

Example: Changing one point

On $[0, 1]$, define $f(x) = 0$ and

$$g(x) = \begin{cases} 1, & x = 1/2, \\ 0, & x \neq 1/2. \end{cases}$$

Then $f = g$ a.e. because they differ only on the singleton $\{1/2\}$.

Counterexample: pointwise equality is stronger

Counterexample: Do not confuse equality with a.e. equality

The functions

$$f(x) = 0, \quad g(x) = \mathbf{1}_{\mathbb{Q} \cap [0,1]}(x)$$

are not equal pointwise. But they are equal almost everywhere with respect to Lebesgue measure.

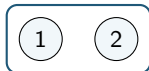
Why?

They differ on $\mathbb{Q} \cap [0, 1]$, a countable set, hence a null set.

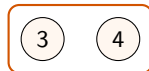
Measurable functions as information

Finite sigma-algebra case

If $X = \{1, 2, 3, 4\}$ and the atoms are $\{1, 2\}$ and $\{3, 4\}$, then a measurable function must be constant on each atom.



$$f = 2$$



$$f = 5$$

Exercise: finite measurable space

Exercise: Measurable or not?

Let $X = \{1, 2, 3, 4\}$ and let $\mathcal{F} = \{\emptyset, X, \{1, 2\}, \{3, 4\}\}$. Define

$$f(1) = 1, \quad f(2) = 2, \quad f(3) = 0, \quad f(4) = 0.$$

Is f measurable?

Solution

No. For example,

$$\{x : f(x) > 1.5\} = \{2\},$$

but $\{2\} \notin \mathcal{F}$. Hence f is not measurable.

Exam-style: limits of measurable functions

Exam-style exercise: Proof task

Let f_n be measurable and suppose $f_n(x) \rightarrow f(x)$ for all x . Prove that f is measurable.

Solution idea

Use the identity

$$f(x) = \limsup_{n \rightarrow \infty} f_n(x) = \inf_{m \geq 1} \sup_{n \geq m} f_n(x).$$

Measurability is preserved by countable suprema and infima. Hence f is measurable.

Module 2

Measurable functions

Measurable functions: definition

Definition: Measurable function

Let $(X, \mathcal{F})$ be a measurable space. A function $f : X \rightarrow \mathbb{R}$ is measurable if

$$\{x \in X : f(x) \in B\} \in \mathcal{F}$$

for every Borel set $B \in \mathcal{B}(\mathbb{R})$.

- Equivalently, $f^{-1}(B) \in \mathcal{F}$ for every Borel set B .
- Measurability means that the function is compatible with the measurable structure.

Useful tests for measurability

Theorem: Half-line tests

A function $f : X \rightarrow \mathbb{R}$ is measurable if and only if any one of the following classes is measurable for every $a \in \mathbb{R}$:

$$\{f < a\}, \quad \{f \leq a\}, \quad \{f > a\}, \quad \{f \geq a\}.$$

Proof idea: Why

Each of these half-line classes generates $\mathcal{B}(\mathbb{R})$.

Examples of measurable functions

Example: Basic examples

- Constant functions are measurable.
- Indicator functions $\mathbf{1}_A$ are measurable if and only if A is measurable.
- Continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ are Borel measurable.
- Step functions with measurable pieces are measurable.

Indicator functions

Theorem: Measurability of indicators

For a set $A \subseteq X$,

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Then $\mathbf{1}_A$ is measurable if and only if $A \in \mathcal{F}$.

Proof idea: Proof

If $\mathbf{1}_A$ is measurable, then

$$A = \{x : \mathbf{1}_A(x) > 1/2\} \in \mathcal{F}.$$

Conversely, preimages of Borel sets under $\mathbf{1}_A$ are among $\emptyset, A, A^c, X$.

Operations preserve measurability

Theorem: Algebra of measurable functions

If f, g are measurable and $a, b \in \mathbb{R}$, then:

$$af + bg, fg, |f|, \max(f, g), \min(f, g)$$

are measurable whenever the expressions are finite.

- Quotients are measurable on the set where the denominator is nonzero.
- Composition $\varphi \circ f$ is measurable if f is measurable and φ is Borel measurable.

Limits preserve measurability

Theorem: Pointwise limits

If f_n are measurable, then the following functions are measurable:

$$\sup_n f_n, \quad \inf_n f_n, \quad \limsup_n f_n, \quad \liminf_n f_n.$$

If $f_n(x) \rightarrow f(x)$ pointwise, then f is measurable.

Proof idea: Key identity

$$\{\sup_n f_n > a\} = \bigcup_{n=1}^{\infty} \{f_n > a\}.$$

Simple functions

Definition: Simple function

A measurable function $s : X \rightarrow \mathbb{R}$ is simple if it takes only finitely many values:

$$s = \sum_{k=1}^m a_k \mathbf{1}_{A_k},$$

where $A_k \in \mathcal{F}$ and $a_k \in \mathbb{R}$.

- Simple functions are the building blocks of Lebesgue integration.
- They play the role of rectangles in Riemann integration.

Example: writing a simple function

Example: Piecewise constant function

Let

$$s(x) = \begin{cases} 2, & 0 \leq x < 1, \\ -1, & 1 \leq x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$s = 2\mathbf{1}_{[0,1)} - \mathbf{1}_{[1,3)}.$$

Exercise: Change representation

Write the same s using the disjoint sets $[0, 1)$, $[1, 3)$ and $\mathbb{R} \setminus [0, 3)$.

Approximation by simple functions

Theorem: Simple approximation

If $f : X \rightarrow [0, \infty]$ is measurable, then there exist nonnegative simple functions s_n such that

$$0 \leq s_n \uparrow f.$$

Proof idea: Construction idea

Quantize the range in steps of size 2^{-n} and truncate above at level n .

Example: quantization

Example: Approximating $f(x) = x$ on $[0, 1]$

Define

$$s_n(x) = \frac{k}{2^n} \quad \text{if} \quad \frac{k}{2^n} \leq x < \frac{k+1}{2^n}.$$

Then

$$0 \leq s_n(x) \leq x, \quad s_n(x) \uparrow x.$$

Counterexample: pointwise definition can hide measurability

Counterexample: Non-measurable indicator

If $V \subset [0, 1]$ is non-measurable, then

$$\mathbf{1}_V$$

is not measurable.

- The issue is not the values 0 and 1.
- The issue is the set on which the value 1 is taken.

Measurability and continuity

Theorem: Continuous implies Borel measurable

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then f is Borel measurable.

Proof idea: Proof

For every open set $O \subseteq \mathbb{R}$, continuity gives $f^{-1}(O)$ open. Since open sets are Borel, preimages of open sets are Borel.

Counterexample: Converse false

The function $\mathbf{1}_{[0,1]}$ is Borel measurable but not continuous.

Exercise: measurable operations

Exercise: Check measurability

Let f, g be measurable. Prove that the following are measurable:

- ① $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$.
- ② $|f|$.
- ③ $\{x : f(x) < g(x)\}$.

Hint

Use $f - g$ and half-line tests.

Exam-style exercise: preimage test

Exam-style exercise: Function on a finite space

Let $X = \{1, 2, 3, 4\}$ and

$$\mathcal{F} = \{\emptyset, \{1, 2\}, \{3, 4\}, X\}.$$

Let $f(1) = f(2) = 0$, $f(3) = 1$, $f(4) = 2$. Is f measurable?

Solution

No. The set $\{f = 1\} = \{3\}$ is not in $\mathcal{F}$.

Exam-style exercise: pointwise limit

Exam-style exercise: Limit of measurable functions

Let f_n be measurable and suppose $f_n \rightarrow f$ pointwise. Prove that f is measurable.

Solution skeleton

Use

$$f = \limsup_{n \rightarrow \infty} f_n = \inf_{m \geq 1} \sup_{n \geq m} f_n.$$

Suprema and infima of measurable functions are measurable.

Lebesgue integral: build from simple functions

simple functions → nonnegative measurable functions → integrable signed functions

Strategy

Define the integral exactly on step functions, extend by monotone approximation, then split a signed function into positive and negative parts.

Integral of a simple function: detailed computation

Example: Step function

Let

$$s = 2\mathbf{1}_{[0,1]} + 5\mathbf{1}_{(1,3]} - \mathbf{1}_{(3,4]}.$$

Then

$$\begin{aligned}\int s \, d\lambda &= 2\lambda([0, 1]) + 5\lambda((1, 3]) - 1\lambda((3, 4]) \\ &= 2 \cdot 1 + 5 \cdot 2 - 1 \cdot 1 = 11.\end{aligned}$$

Positive and negative parts

Definition: Decomposition

For a real-valued function f , define

$$f^+ = \max(f, 0), \quad f^- = \max(-f, 0).$$

Then

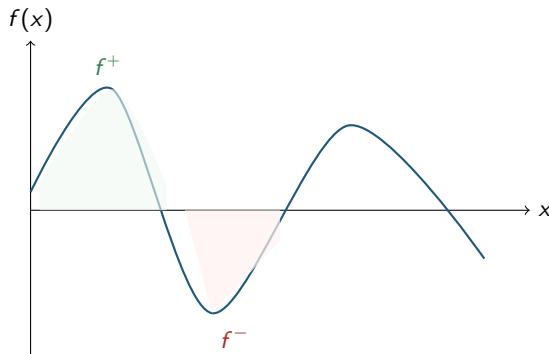
$$f = f^+ - f^-, \quad |f| = f^+ + f^-.$$

Integral of signed functions

If at least one of $\int f^+ d\mu$, $\int f^- d\mu$ is finite, define

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu.$$

Visual: positive and negative parts



Intuition

The signed integral is area above the axis minus area below the axis.

Exercise: integrability of a power near zero

Exercise: Improper-looking but Lebesgue integrable?

For which $a > 0$ is

$$f(x) = x^{-a} \mathbf{1}_{(0,1)}(x)$$

Lebesgue integrable on $(0, 1)$?

Solution

Compute

$$\int_0^1 x^{-a} dx = \begin{cases} \frac{1}{1-a}, & a < 1, \\ +\infty, & a \geq 1. \end{cases}$$

Thus $f \in L^1(0, 1)$ if and only if $a < 1$.

Exercise: integrability at infinity

Exercise: Tail behavior

For which $a > 0$ is

$$f(x) = x^{-a} \mathbf{1}_{(1,\infty)}(x)$$

integrable on $(1, \infty)$?

Solution

Compute

$$\int_1^\infty x^{-a} dx$$

which is finite if and only if $a > 1$. Thus $f \in L^1(1, \infty)$ exactly when $a > 1$.

Exercise: same formula, different spaces

Exercise: Local versus global

Consider $f(x) = 1/x$. Decide whether $f \in L^1(0, 1)$ and whether $f \in L^1(1, \infty)$.

Solution

Near zero,

$$\int_0^1 \frac{1}{x} dx = \infty.$$

At infinity,

$$\int_1^\infty \frac{1}{x} dx = \infty.$$

Thus $1/x$ is not integrable on either interval.

The Dirichlet function revisited

Example: Lebesgue integral of a highly discontinuous function

Let

$$f = \mathbf{1}_{\mathbb{Q} \cap [0,1]}.$$

Since $\mathbb{Q} \cap [0,1]$ is countable, it has measure zero. Hence

$$\int_0^1 f \, d\lambda = \lambda(\mathbb{Q} \cap [0,1]) = 0.$$

Moral

The Lebesgue integral is sensitive to the size of the set, not to the number of discontinuities.

Functions equal a.e. have the same integral

Theorem: Null changes do not matter

If $f = g$ almost everywhere and both are integrable, then

$$\int f \, d\mu = \int g \, d\mu.$$

Example: One-point modification

On $[0, 1]$, changing the value of a function at finitely many points does not change its Lebesgue integral.

Exercise: a.e. equality and integral

Exercise: Compute without integrating

Let

$$f(x) = x^2, \quad g(x) = \begin{cases} x^2, & x \neq 1/2, \\ 100, & x = 1/2. \end{cases}$$

on $[0, 1]$. Compute $\int_0^1 g(x) dx$.

Solution

The functions differ only at one point. Therefore $f = g$ a.e. and

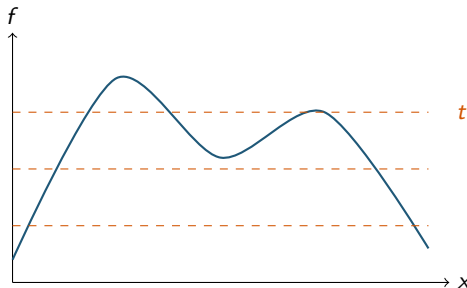
$$\int_0^1 g(x) dx = \int_0^1 x^2 dx = \frac{1}{3}.$$

Layer-cake intuition

Nonnegative functions can be integrated by levels

For $f \geq 0$, the integral can be understood as summing the sizes of the level sets

$$\{x : f(x) > t\}.$$



Exam-style: integral of an indicator combination

Exam-style exercise: Calculation

Let A, B be measurable sets with

$$\mu(A) = 5, \quad \mu(B) = 7, \quad \mu(A \cap B) = 2.$$

Compute

$$\int (3\mathbf{1}_A + 4\mathbf{1}_B) d\mu.$$

Solution

Use linearity:

$$\int (3\mathbf{1}_A + 4\mathbf{1}_B) d\mu = 3\mu(A) + 4\mu(B) = 15 + 28 = 43.$$

The intersection is irrelevant here because the expression is a sum of indicators, not an indicator of a union.

Module 2

Lebesgue integration

Integral of a nonnegative simple function

Definition: Simple integral

Let

$$s = \sum_{k=1}^m a_k \mathbf{1}_{A_k}, \quad a_k \geq 0,$$

with A_k disjoint. Define

$$\int_X s \, d\mu = \sum_{k=1}^m a_k \mu(A_k).$$

- The value is independent of the chosen disjoint representation.
- Infinite values are allowed.

Example: simple integral

Example: Step function

Let $s = 3\mathbf{1}_{[0,2]} + 5\mathbf{1}_{(2,4]}$ on $\mathbb{R}$ with Lebesgue measure. Then

$$\int s \, d\lambda = 3\lambda([0, 2]) + 5\lambda((2, 4]) = 3 \cdot 2 + 5 \cdot 2 = 16.$$

Exercise: Compute

Evaluate $\int (2\mathbf{1}_{[0,1]} - \mathbf{1}_{[1,3]}) \, d\lambda$.

Integral of a nonnegative measurable function

Definition: Lebesgue integral for $f \geq 0$

For measurable $f : X \rightarrow [0, \infty]$, define

$$\int_X f \, d\mu = \sup \left\{ \int_X s \, d\mu : 0 \leq s \leq f, \, s \text{ simple} \right\}.$$

- The integral may be $+\infty$.
- It extends the simple integral.

Integral of a signed function

Definition: Positive and negative parts

For a real measurable function f , define

$$f^+ = \max(f, 0), \quad f^- = \max(-f, 0).$$

Then

$$f = f^+ - f^-, \quad |f| = f^+ + f^-.$$

Definition: Integrable function

f is integrable if

$$\int |f| \, d\mu < \infty.$$

Then

$$\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu.$$

Basic properties of the integral

Theorem: Linearity and monotonicity

If f, g are integrable and $a, b \in \mathbb{R}$, then

$$\int (af + bg) d\mu = a \int f d\mu + b \int g d\mu.$$

If $f \leq g$ a.e., then

$$\int f d\mu \leq \int g d\mu.$$

Also

$$\left| \int f d\mu \right| \leq \int |f| d\mu.$$

Integral over a set

Definition: Restricted integral

For $A \in \mathcal{F}$ define

$$\int_A f \, d\mu = \int_X f \mathbf{1}_A \, d\mu.$$

Example: Integral of an indicator

$$\int_X \mathbf{1}_A \, d\mu = \mu(A).$$

Exercise: Immediate computation

Compute $\int_{[1,4]} x \, dx$ using the usual formula and interpret it as a Lebesgue integral.

Relation with the Riemann integral

Theorem: Compatibility

If $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable, then it is Lebesgue integrable and the two integrals coincide.

Meaning

Lebesgue integration extends the usual integral, but keeps the same values for well-behaved functions.

Example: Classical integral

$$\int_0^1 x^2 d\lambda = \frac{1}{3}.$$

Example: Dirichlet function

Example: Everywhere discontinuous but integrable

Let

$$f = \mathbf{1}_{\mathbb{Q} \cap [0,1]}.$$

Then $f = 0$ a.e. and

$$\int_0^1 f \, d\lambda = 0.$$

- This function is not Riemann integrable.
- Lebesgue integration sees that the nonzero set has measure zero.

Functions equal a.e. have same integral

Theorem: A.e. invariance

If $f = g$ a.e. and f, g are integrable, then

$$\int f \, d\mu = \int g \, d\mu.$$

Proof idea: Proof idea

Since $|f - g| = 0$ a.e.,

$$\int |f - g| \, d\mu = 0.$$

Then use

$$\left| \int f - \int g \right| \leq \int |f - g| \, d\mu.$$

When integral zero implies function zero

Theorem: Nonnegative case

If $f \geq 0$ and

$$\int f \, d\mu = 0,$$

then $f = 0$ a.e.

Proof idea: Proof idea

For $n \geq 1$, define $A_n = \{f \geq 1/n\}$. Then

$$0 = \int f \, d\mu \geq \frac{1}{n} \mu(A_n),$$

so $\mu(A_n) = 0$. Since $\{f > 0\} = \bigcup_n A_n$, the claim follows.

Counterexample: signed integral zero

Counterexample: Integral zero does not imply function zero

On $[-1, 1]$ let $f(x) = x$. Then

$$\int_{-1}^1 x \, dx = 0,$$

but $f \neq 0$ a.e.

- The theorem requires $f \geq 0$.
- Positive and negative areas can cancel.

Integrability examples

Example: Power functions near zero

For $a \in \mathbb{R}$,

$$\int_0^1 x^{-a} dx < \infty \iff a < 1.$$

Example: Power functions at infinity

$$\int_1^\infty x^{-a} dx < \infty \iff a > 1.$$

- Singularities and tails require different tests.

Exercise: decide integrability

Exercise: Integrability tests

Decide whether the following functions are integrable with respect to Lebesgue measure:

- ① $f(x) = 1/\sqrt{x}$ on $(0, 1)$.
- ② $g(x) = 1/x$ on $(0, 1)$.
- ③ $h(x) = 1/(1 + x^2)$ on $\mathbb{R}$.
- ④ $k(x) = 1/(1 + |x|)$ on $\mathbb{R}$.

Answers

Integrable: f, h . Not integrable: g, k .

Markov-type inequality

Theorem: Integral controls large values

If $f \geq 0$ is integrable, then for every $a > 0$,

$$\mu(\{f \geq a\}) \leq \frac{1}{a} \int f \, d\mu.$$

Proof idea: One-line proof

Since $f \geq a \mathbf{1}_{\{f \geq a\}}$,

$$\int f \, d\mu \geq a \mu(\{f \geq a\}).$$

Practical use: bounding exceptional sets

Example: From average size to set size

If $f \geq 0$ and $\int f d\mu \leq 0.02$, then

$$\mu(\{f \geq 0.1\}) \leq \frac{0.02}{0.1} = 0.2.$$

- Integral bounds give information on where a function can be large.
- This idea appears repeatedly in analysis.

Change of value on a null set

Example: Modifying a function at one point

Let

$$f(x) = 0 \quad \text{for all } x,$$

and define $g(0) = 1000$, $g(x) = 0$ for $x \neq 0$. Then

$$f = g \quad \text{a.e.,} \quad \int g \, d\lambda = 0.$$

Moral

Lebesgue integration ignores null-set modifications.

Exam-style exercise: integral zero

Exam-style exercise: Proof question

Let $f \geq 0$ be measurable. Prove that

$$\int f \, d\mu = 0 \iff f = 0 \text{ a.e.}$$

Solution

The nontrivial implication uses $A_n = \{f \geq 1/n\}$ and

$$\{f > 0\} = \bigcup_{n=1}^{\infty} A_n.$$

Exam-style exercise: simple integral

Exam-style exercise: Computation

Let $X = \{1, 2, 3, 4\}$, $\mu(\{1\}) = 1$, $\mu(\{2\}) = 2$, $\mu(\{3\}) = 0$, $\mu(\{4\}) = 3$. If

$$f(1) = 4, \quad f(2) = -1, \quad f(3) = 100, \quad f(4) = 2,$$

compute $\int f \, d\mu$.

Solution

$$4 \cdot 1 + (-1) \cdot 2 + 100 \cdot 0 + 2 \cdot 3 = 8.$$

Why convergence theorems are needed

The question

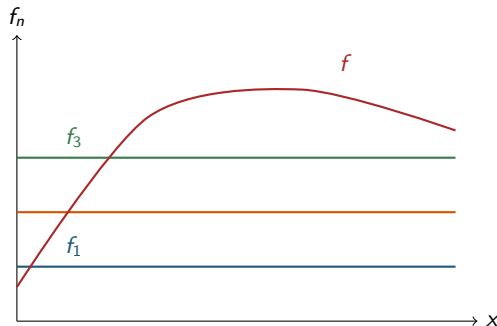
When is it legitimate to write

$$\int \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu?$$

Counterexample: Warning

Pointwise convergence alone is not enough. The moving spike example has $f_n \rightarrow 0$ pointwise but $\int f_n = 1$.

Monotone Convergence Theorem: intuition



Theorem: MCT

If $0 \leq f_n \uparrow f$, then

$$\int f_n d\mu \uparrow \int f d\mu.$$

Exercise: apply MCT

Exercise: Increasing indicators

Let $f_n = \mathbf{1}_{[0, 1 - 1/n]}$ on $[0, 1]$. Find $\lim_n \int f_n d\lambda$.

Solution

The sequence increases pointwise to $\mathbf{1}_{[0, 1]}$. By MCT,

$$\lim_n \int f_n d\lambda = \int \mathbf{1}_{[0, 1]} d\lambda = 1.$$

Directly, $\int f_n = 1 - 1/n \rightarrow 1$.

Fatou lemma: safety inequality

Theorem: Fatou

If $f_n \geq 0$, then

$$\int \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu.$$

Interpretation

Fatou gives an inequality even when neither monotonicity nor domination is available.

Fatou in the moving spike example

Example: Moving spike

For $f_n = n\mathbf{1}_{(0,1/n)}$ on $[0, 1]$, we have $f_n \rightarrow 0$ pointwise, so

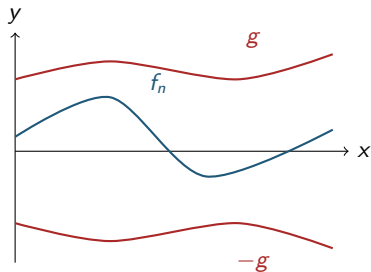
$$\int \liminf f_n = 0.$$

But

$$\liminf \int f_n = 1.$$

Fatou says $0 \leq 1$, which is true but not equality.

Dominated Convergence Theorem: intuition



Theorem: DCT

If $f_n \rightarrow f$ a.e. and $|f_n| \leq g$ with $g \in L^1$, then

$$\int f_n d\mu \rightarrow \int f d\mu.$$

Image

All functions stay inside an integrable envelope.

Exercise: apply DCT

Exercise: A standard limit

Compute

$$\lim_{n \rightarrow \infty} \int_0^1 x^n dx$$

using a convergence theorem.

Solution

For $x \in [0, 1)$, $x^n \rightarrow 0$, and at $x = 1$, $x^n = 1$. Hence $x^n \rightarrow 0$ a.e. Moreover $0 \leq x^n \leq 1$, and $1 \in L^1([0, 1])$. By DCT,

$$\int_0^1 x^n dx \rightarrow 0.$$

Exercise: DCT fails without domination

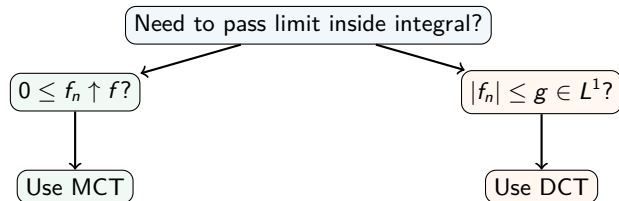
Exercise: Find the missing hypothesis

Let $f_n = n\mathbf{1}_{(0,1/n)}$ on $[0, 1]$. We know $f_n \rightarrow 0$ a.e. Why can DCT not be applied?

Solution

There is no integrable function g such that $|f_n| \leq g$ for all n . If such g existed, it would have to dominate arbitrarily high spikes near zero. The integrals also reveal the problem: $\int f_n = 1$ does not converge to 0.

MCT versus DCT: decision tree



If neither applies

Try Fatou for an inequality or look for a counterexample.

Exam-style: choose the correct theorem

Exam-style exercise: Three sequences

On $[0, 1]$, identify a suitable theorem for each sequence:

- ① $f_n(x) = \mathbf{1}_{[0, 1-1/n]}(x)$.
- ② $g_n(x) = x^n$.
- ③ $h_n(x) = n\mathbf{1}_{(0, 1/n)}(x)$.

Solution

(1) MCT: increasing indicators. (2) DCT: dominated by 1, converges a.e. to 0. (3) No equality theorem applies; Fatou gives only an inequality.

Counterexample: convergence in measure is not enough alone

Counterexample: Another moving spike

The sequence $f_n = n\mathbf{1}_{(0,1/n)}$ converges to 0 in measure on $[0, 1]$, but

$$\int_0^1 f_n d\lambda = 1$$

for every n .

Lesson

Modes of convergence must be matched with hypotheses controlling mass.

Exam-style proof: DCT implies L1 convergence under domination

Exam-style exercise: Proof task

Assume $f_n \rightarrow f$ a.e. and $|f_n| \leq g \in L^1$. Prove that $\|f_n - f\|_1 \rightarrow 0$.

Solution

Since $f_n \rightarrow f$ a.e., $|f_n - f| \rightarrow 0$ a.e. Also

$$|f_n - f| \leq |f_n| + |f|.$$

Because $|f| \leq g$ a.e., we get $|f_n - f| \leq 2g$, and $2g \in L^1$. By DCT,

$$\int |f_n - f| d\mu \rightarrow 0.$$

Module 2

Convergence theorems

Why convergence theorems matter

The central question

When is it legitimate to write

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int \lim_{n \rightarrow \infty} f_n d\mu?$$

- It is not always true.
- Measure theory gives powerful sufficient conditions.
- The three main tools are MCT, Fatou and DCT.

Monotone Convergence Theorem

Theorem: MCT

Let $0 \leq f_1 \leq f_2 \leq \dots$ be measurable and suppose

$$f_n(x) \uparrow f(x) \quad \text{for all } x.$$

Then

$$\int f_n d\mu \uparrow \int f d\mu.$$

- No boundedness assumption is needed.
- The limit may be $+\infty$.

MCT example: expanding intervals

Example: Indicator sequence

Let $f_n = \mathbf{1}_{[0,n]}$ on $\mathbb{R}$. Then $f_n \uparrow 1$ pointwise and

$$\int f_n d\lambda = n \uparrow \infty = \int 1 d\lambda.$$

- The theorem permits infinite limits.

MCT example: power series under the integral

Example: Exponential series on a finite interval

For $x \in [0, 1]$,

$$s_n(x) = \sum_{k=0}^n \frac{x^k}{k!} \uparrow e^x.$$

By MCT,

$$\int_0^1 e^x dx = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{1}{k!(k+1)}.$$

Proof idea of MCT

Proof idea: Key mechanism

The inequality

$$\lim_n \int f_n d\mu \leq \int f d\mu$$

follows from monotonicity. For the reverse inequality, approximate f from below by a simple function $s \leq f$ and use the sets where f_n is close to f .

Take-away

The proof relies on simple-function approximation.

Fatou Lemma

Theorem: Fatou

If $f_n \geq 0$ are measurable, then

$$\int \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu.$$

- Fatou gives an inequality, not generally equality.
- It is often used to prove lower semicontinuity.

Proof idea of Fatou

Proof idea: From MCT

Set

$$g_n = \inf_{k \geq n} f_k.$$

Then $g_n \uparrow \liminf f_n$. By MCT,

$$\int \liminf f_n = \lim_n \int g_n.$$

Since $g_n \leq f_k$ for all $k \geq n$, we get

$$\int g_n \leq \inf_{k \geq n} \int f_k.$$

Take limits.

Fatou can be strict

Counterexample: Strict inequality

Let $f_n = \mathbf{1}_{[n, n+1]}$ on $\mathbb{R}$. Then $f_n(x) \rightarrow 0$ for every x , so

$$\int \liminf f_n d\lambda = 0.$$

But

$$\int f_n d\lambda = 1$$

for all n , hence the right side is 1.

Dominated Convergence Theorem

Theorem: DCT

Let $f_n \rightarrow f$ pointwise a.e. Suppose there exists an integrable function g such that

$$|f_n| \leq g \quad \text{a.e. for all } n.$$

Then f is integrable and

$$\int f_n d\mu \rightarrow \int f d\mu.$$

Moreover,

$$\int |f_n - f| d\mu \rightarrow 0.$$

DCT example: bounded oscillations

Example: Oscillatory sequence

On $[0, 1]$, let

$$f_n(x) = x^2 \sin(nx).$$

Then $|f_n(x)| \leq x^2 \leq 1$, and 1 is integrable on $[0, 1]$.

- If f_n converges pointwise to 0 along a sequence, DCT yields $\int f_n \rightarrow 0$.
- The domination is the important part.

DCT example: parameter limit

Example: Limit inside integral

Compute

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{x^n}{1+x} dx.$$

For $x \in [0, 1)$, $x^n/(1+x) \rightarrow 0$; at $x = 1$ the limit is $1/2$. The point $x = 1$ has measure zero. Also

$$0 \leq \frac{x^n}{1+x} \leq 1.$$

Hence the limit is 0.

DCT fails without domination: moving mass

Counterexample: No integrable dominating function

Let

$$f_n = n \mathbf{1}_{(0,1/n)}.$$

Then $f_n(x) \rightarrow 0$ for every $x > 0$, but

$$\int_0^1 f_n(x) dx = 1$$

for all n .

- Pointwise convergence alone is not enough.
- Mass concentrates near zero.

DCT fails without finite domination: escaping mass

Counterexample: Mass goes to infinity

On $\mathbb{R}$, let

$$f_n = \mathbf{1}_{[n, n+1]}.$$

Then $f_n \rightarrow 0$ pointwise, but

$$\int f_n d\lambda = 1.$$

- No integrable function can dominate all f_n .
- The mass escapes to infinity.

How to choose the theorem

Decision tree

- ① Are the functions nonnegative and increasing? Use MCT.
- ② Are they nonnegative without monotonicity? Fatou gives an inequality.
- ③ Do they converge pointwise and have an integrable common bound? Use DCT.
- ④ If none applies, look for a counterexample or a direct estimate.

Exercise: identify the theorem

Exercise: MCT, Fatou or DCT?

For each sequence, decide which theorem applies on $(0, 1)$:

- ① $f_n = x^n$.
- ② $f_n = n\mathbf{1}_{(0, 1/n)}$.
- ③ $f_n = \sum_{k=1}^n 2^{-k}\mathbf{1}_{(0, 1/k)}$.

Answers

1: DCT. 2: Fatou only gives inequality; DCT fails. 3: MCT.

Uniform convergence versus DCT

Two different ideas

Uniform convergence on a finite-measure set often permits passage of limits through integrals:

$$\left| \int f_n - \int f \right| \leq \mu(X) \sup_x |f_n - f|.$$

DCT is different: it uses pointwise convergence plus domination.

Counterexample: Finite measure is needed

Uniform convergence to zero on $\mathbb{R}$ does not imply integrals go to zero. Example:

$$f_n = \frac{1}{n} \mathbf{1}_{[0, n^2]}.$$

Then $\sup f_n \rightarrow 0$ but $\int f_n = n$.

Exam-style exercise: DCT computation

Exam-style exercise: Limit of integrals

Show that

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{\sin(nx)}{n+x} dx = 0.$$

Solution

For all $x \in [0, 1]$,

$$\left| \frac{\sin(nx)}{n+x} \right| \leq \frac{1}{n} \leq 1.$$

Pointwise convergence to 0 is immediate. DCT with dominating function 1 gives the result.

Exam-style exercise: MCT computation

Exam-style exercise: Series under the integral

Let $a_k \geq 0$ and $g_k \geq 0$ measurable. Prove that

$$\int \sum_{k=1}^{\infty} a_k g_k d\mu = \sum_{k=1}^{\infty} a_k \int g_k d\mu.$$

Solution

Set $s_n = \sum_{k=1}^n a_k g_k$. Then $s_n \uparrow \sum_{k \geq 1} a_k g_k$ and apply MCT.

Exam-style counterexample: DCT hypotheses

Exam-style exercise: Find what fails

Let $f_n = n\mathbf{1}_{(0,1/n)}$ on $(0,1)$. Show that $f_n \rightarrow 0$ a.e. but

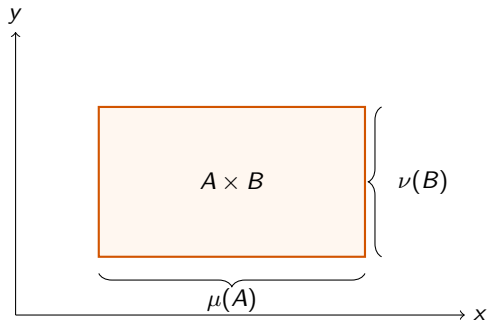
$$\int_0^1 f_n dx \not\rightarrow 0.$$

Which hypothesis of DCT fails?

Answer

The missing hypothesis is the existence of one integrable function g with $|f_n| \leq g$ for all n .

Product measure: visual intuition



Rectangle rule

Product measure satisfies

$$(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B).$$

Tonelli versus Fubini

Tonelli

For $f \geq 0$, iterated integrals and double integral agree, possibly with value $+\infty$.

Fubini

For integrable f , iterated integrals and double integral agree and are finite.

Counterexample: Warning

If f changes sign and is not integrable, exchanging order can be dangerous.

Exercise: rectangle integral

Exercise: Product of one-variable functions

Compute

$$\int_0^1 \int_0^2 xy \, dy \, dx.$$

Solution

First integrate in y :

$$\int_0^2 xy \, dy = x \left[\frac{y^2}{2} \right]_0^2 = 2x.$$

Then integrate in x :

$$\int_0^1 2x \, dx = 1.$$

Exercise: change order of integration

Exercise: Triangle

Compute

$$\int_0^1 \int_0^x (x + y) dy dx.$$

Solution

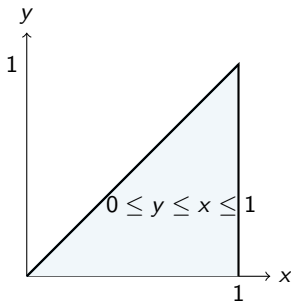
The region is $0 \leq y \leq x \leq 1$. Compute directly:

$$\int_0^x (x + y) dy = x^2 + \frac{y^2}{2} \Big|_0^x = \frac{3}{2}x^2.$$

Thus

$$\int_0^1 \frac{3}{2}x^2 dx = \frac{1}{2}.$$

Visual: triangular region



Changing order

The same region can be written as

$$0 \leq y \leq 1, \quad y \leq x \leq 1.$$

Exercise: Check

Draw a horizontal slice and identify its x -limits.

Exam-style: use Fubini to factor integrals

Exam-style exercise: Product function

Let $f \in L^1(X)$ and $g \in L^1(Y)$. Show that

$$\int_{X \times Y} f(x)g(y) d(\mu \otimes \nu) = \left(\int_X f d\mu \right) \left(\int_Y g d\nu \right).$$

Solution

By Fubini,

$$\int_X \left(\int_Y f(x)g(y) d\nu(y) \right) d\mu(x) = \int_X f(x) \left(\int_Y g(y) d\nu(y) \right) d\mu(x).$$

The inner integral is constant in x , giving the product.

Counterexample mindset: signs matter

Counterexample: What can go wrong

There are functions for which

$$\int \left(\int f(x, y) dy \right) dx$$

and

$$\int \left(\int f(x, y) dx \right) dy$$

exist but give different values, because f is not absolutely integrable.

Exam lesson

Before using Fubini for signed functions, check integrability of $|f|$ or quote the correct hypothesis.

Module 2

Product measures and Fubini-Tonelli

Why products appear

Two variables

Many integrals involve functions of two variables:

$$\int \int f(x, y) d\mu(x) d\nu(y).$$

We need a rigorous way to integrate on $X \times Y$.

- Product sigma-algebra.
- Product measure.
- Tonelli theorem for nonnegative functions.
- Fubini theorem for integrable functions.

Product sigma-algebra

Definition: Product measurable space

Given $(X, \mathcal{F})$ and $(Y, \mathcal{G})$, define

$$\mathcal{F} \otimes \mathcal{G} = \sigma(\{A \times B : A \in \mathcal{F}, B \in \mathcal{G}\}).$$

- Measurable rectangles generate the product structure.
- On $\mathbb{R}^2$, $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) = \mathcal{B}(\mathbb{R}^2)$.

Product measure on rectangles

Definition: Product measure

For sigma-finite measure spaces, there exists a unique product measure $\mu \otimes \nu$ satisfying

$$(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B).$$

Example: Lebesgue measure in the plane

Area is the product of lengths on rectangles:

$$\lambda^2([a, b] \times [c, d]) = (b - a)(d - c).$$

Tonelli theorem

Theorem: Tonelli

If $f : X \times Y \rightarrow [0, \infty]$ is measurable, then

$$\int_{X \times Y} f \, d(\mu \otimes \nu) = \int_X \left(\int_Y f(x, y) \, d\nu(y) \right) d\mu(x),$$

and also with the order reversed.

- Nonnegative functions do not require integrability assumptions.

Fubini theorem

Theorem: Fubini

If f is integrable on $X \times Y$, i.e.

$$\int_{X \times Y} |f| d(\mu \otimes \nu) < \infty,$$

then the iterated integrals exist a.e. and

$$\int f d(\mu \otimes \nu) = \int_X \int_Y f(x, y) d\nu(y) d\mu(x) = \int_Y \int_X f(x, y) d\mu(x) d\nu(y).$$

Example: computing a double integral

Example: Triangle integral

Compute

$$\int_0^1 \int_0^x y \, dy \, dx.$$

Inner integral:

$$\int_0^x y \, dy = \frac{x^2}{2}.$$

Thus

$$\int_0^1 \frac{x^2}{2} \, dx = \frac{1}{6}.$$

Exercise: Reverse the order

Rewrite the same integral as $\int_0^1 \int_y^1 y \, dx \, dy$ and compute it.

Counterexample: order matters without integrability

Counterexample: Conditional cancellation

There exist functions f on $\mathbb{R}^2$ such that the two iterated integrals exist but are different, because

$$\int |f| = \infty.$$

Practical rule

Before swapping integrals for signed functions, check integrability of $|f|$, or use Tonelli when $f \geq 0$.

Exam-style exercise: Tonelli or Fubini?

Exam-style exercise: Choose the theorem

Let

$$f(x, y) = e^{-x} e^{-y} \mathbf{1}_{x \geq 0} \mathbf{1}_{y \geq 0}.$$

Compute $\int_{\mathbb{R}^2} f(x, y) \, dx dy$.

Solution

Since $f \geq 0$, Tonelli applies:

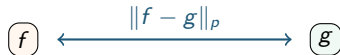
$$\int_0^\infty e^{-x} dx \int_0^\infty e^{-y} dy = 1.$$

L^p spaces: the big picture

From functions to geometry

In L^p , functions are points and the distance between them is measured by the size of their difference:

$$\|f - g\|_p = \left(\int |f - g|^p d\mu \right)^{1/p}.$$



Do not forget equivalence classes

Important correction

Strictly speaking, an element of L^p is not a single function, but an equivalence class of functions equal almost everywhere.

Example: Same L^p element

On $[0, 1]$, the functions

$$f(x) = 0, \quad g(x) = \mathbf{1}_{\{1/2\}}(x)$$

represent the same element of L^p for every $1 \leq p \leq \infty$.

Computing L^p norms: first example

Example: $f(x) = x$ on $[0, 1]$

For $1 \leq p < \infty$,

$$\|x\|_p = \left(\int_0^1 x^p dx \right)^{1/p} = \left(\frac{1}{p+1} \right)^{1/p}.$$

In particular,

$$\|x\|_1 = \frac{1}{2}, \quad \|x\|_2 = \frac{1}{\sqrt{3}}.$$

Exercise: norm of an indicator

Exercise: Indicator function

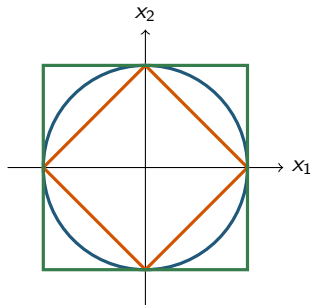
Let A be measurable with $\mu(A) < \infty$. Compute $\|\mathbf{1}_A\|_p$ for $1 \leq p < \infty$.

Solution

Since $|\mathbf{1}_A|^p = \mathbf{1}_A$,

$$\|\mathbf{1}_A\|_p = \left(\int \mathbf{1}_A d\mu \right)^{1/p} = \mu(A)^{1/p}.$$

Unit balls in different norms



Legend

- orange: $p = 1$;
- blue: $p = 2$;
- green: $p = \infty$.

Intuition

The parameter p changes what it means for a vector or function to be large.

Inclusions depend on the measure space

Theorem: Finite measure case

If $\mu(X) < \infty$ and $1 \leq q < p \leq \infty$, then

$$L^p(X) \subset L^q(X).$$

Counterexample: Infinite measure warning

On infinite measure spaces, this inclusion can fail.

Exercise: L^p membership on $(0, 1)$

Exercise: Power singularity

For which $p \geq 1$ does

$$f(x) = x^{-a}$$

belong to $L^p(0, 1)$?

Solution

We need

$$\int_0^1 |x^{-a}|^p dx = \int_0^1 x^{-ap} dx < \infty.$$

This holds if and only if $ap < 1$. Therefore

$$f \in L^p(0, 1) \iff p < \frac{1}{a}.$$

Exercise: L^p membership on $(1, \infty)$

Exercise: Power tail

For which $p \geq 1$ does

$$f(x) = x^{-a}$$

belong to $L^p(1, \infty)$?

Solution

We need

$$\int_1^\infty x^{-ap} dx < \infty.$$

This holds if and only if $ap > 1$. Hence

$$f \in L^p(1, \infty) \iff p > \frac{1}{a}.$$

Holder inequality: intuition

Theorem: Holder

If $1 < p, q < \infty$ and $1/p + 1/q = 1$, then

$$\int |fg| d\mu \leq \|f\|_p \|g\|_q.$$

Interpretation

The product fg is integrable if one factor is controlled in L^p and the other in the conjugate space L^q .

Exercise: use Holder

Exercise: Bounding an integral

Let $f \in L^2(0, 1)$. Show that $f \in L^1(0, 1)$ and

$$\|f\|_1 \leq \|f\|_2.$$

Solution

Use Holder with $g = 1$, $p = q = 2$:

$$\int_0^1 |f| dx \leq \left(\int_0^1 |f|^2 dx \right)^{1/2} \left(\int_0^1 1^2 dx \right)^{1/2} = \|f\|_2.$$

Minkowski inequality: triangle inequality in L^p

Theorem: Minkowski

For $1 \leq p < \infty$,

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Meaning

This is the triangle inequality that turns L^p into a normed space.

Exercise: Quick check

Verify Minkowski directly for constant functions on $[0, 1]$.

L^2 and inner products

Definition: L^2 inner product

For real-valued functions,

$$\langle f, g \rangle = \int fg \, d\mu.$$

Then

$$\|f\|_2 = \sqrt{\langle f, f \rangle}.$$

Geometry

L^2 is the space where we can speak about angles, orthogonality and projections.

Exercise: orthogonality in L^2

Exercise: Two functions on $[-1, 1]$

Show that $f(x) = 1$ and $g(x) = x$ are orthogonal in $L^2([-1, 1])$.

Solution

Compute the inner product:

$$\langle f, g \rangle = \int_{-1}^1 1 \cdot x \, dx = \left[\frac{x^2}{2} \right]_{-1}^1 = 0.$$

Thus $f \perp g$.

Best constant approximation in L^2

Exam-style exercise: Projection exercise

Find the constant c minimizing

$$\int_0^1 (x - c)^2 dx.$$

Solution

Differentiate:

$$\frac{d}{dc} \int_0^1 (x - c)^2 dx = -2 \int_0^1 (x - c) dx.$$

Set this equal to zero:

$$\int_0^1 x dx - c \int_0^1 dx = 0 \quad \Rightarrow \quad c = \frac{1}{2}.$$

This is the L^2 projection of x onto the constants.

Completeness of L^p : why it matters

Theorem: Riesz-Fischer idea

For $1 \leq p \leq \infty$, L^p is complete.

Interpretation

If a sequence of functions is Cauchy in L^p , then it converges in L^p to a function that still belongs to L^p .

Exercise: Conceptual question

Why would incompleteness be problematic for approximation methods?

Convergence in L^p versus a.e. convergence

Counterexample: Different notions

Convergence in L^p and almost everywhere convergence are related but not identical.

Example: Moving intervals

There are sequences converging to zero in L^1 but not pointwise everywhere, and sequences converging pointwise but not in L^1 .

Exam strategy

Always identify the mode of convergence before applying a theorem.

Exercise: L^1 convergence

Exercise: Shrinking support

Let $f_n = \mathbf{1}_{(0,1/n)}$ on $[0, 1]$. Does $f_n \rightarrow 0$ in L^1 ?

Solution

Yes:

$$\|f_n - 0\|_1 = \int_0^1 \mathbf{1}_{(0,1/n)} dx = \frac{1}{n} \rightarrow 0.$$

The shrinking support kills the L^1 mass.

Exercise: pointwise but not L^1

Exercise: Spike again

Let $f_n = n\mathbf{1}_{(0,1/n)}$ on $[0, 1]$. Does $f_n \rightarrow 0$ in L^1 ?

Solution

No:

$$\|f_n\|_1 = \int_0^1 n\mathbf{1}_{(0,1/n)} dx = 1.$$

Although $f_n \rightarrow 0$ pointwise, the L^1 distance from zero remains equal to 1.

Essential supremum

Definition: L^∞ norm

The L^∞ norm is

$$\|f\|_\infty = \operatorname{ess\,sup} |f|.$$

It ignores values changed on null sets.

Example: One large value at one point

If $f = 0$ on $[0, 1]$ except $f(1/2) = 1000$, then

$$\|f\|_\infty = 0$$

as an L^∞ element.

Exam-style: norm or not?

Exam-style exercise: Subtle issue

On actual functions, define

$$N(f) = \left(\int |f|^p d\mu \right)^{1/p}.$$

Is N a norm on the set of measurable functions?

Solution

Not on actual functions: $N(f) = 0$ only implies $f = 0$ almost everywhere, not pointwise. It becomes a true norm on equivalence classes modulo a.e. equality. That space is L^p .

Module 2

L^p spaces

From integrals to norms

Definition: L^p class

For $1 \leq p < \infty$, define

$$L^p(X, \mu) = \left\{ f \text{ measurable} : \int_X |f|^p d\mu < \infty \right\},$$

with

$$\|f\|_p = \left(\int_X |f|^p d\mu \right)^{1/p}.$$

- Strictly speaking, L^p consists of equivalence classes modulo a.e. equality.

Definition: Essential supremum

A measurable function f is essentially bounded if there exists $M < \infty$ such that

$$|f(x)| \leq M \quad \text{a.e.}$$

Define

$$\|f\|_\infty = \text{ess sup } |f|.$$

Example: Ignoring null spikes

If $f(0) = 1000$ and $f(x) = 0$ for $x \neq 0$, then

$$\|f\|_\infty = 0$$

with Lebesgue measure.

Why equivalence classes?

Counterexample: A seminorm on functions

If $f = \mathbf{1}_{\{0\}}$ on $\mathbb{R}$, then

$$\|f\|_p = 0$$

for every finite p , but $f \neq 0$ pointwise.

Solution

Identify functions that are equal a.e. Then $\|f\|_p = 0$ implies $f = 0$ in L^p .

Examples of L^p membership

Example: Power tails

On $(1, \infty)$, let

$$f(x) = x^{-a}.$$

Then

$$f \in L^p(1, \infty) \iff ap > 1.$$

Example: Power singularities

On $(0, 1)$, $x^{-a} \in L^p(0, 1)$ iff

$$ap < 1.$$

Exercise: classify membership

Exercise: L^p membership

Decide for which $p \geq 1$ the function belongs to L^p :

- ① $f(x) = 1/x$ on $(1, \infty)$.
- ② $g(x) = 1/\sqrt{x}$ on $(0, 1)$.
- ③ $h(x) = e^{-x}$ on $(0, \infty)$.

Answers

1: $p > 1$. 2: $p < 2$. 3: all $p \geq 1$.

Holder inequality

Theorem: Holder

Let $1 < p, q < \infty$ with

$$\frac{1}{p} + \frac{1}{q} = 1.$$

If $f \in L^p$ and $g \in L^q$, then $fg \in L^1$ and

$$\int |fg| \, d\mu \leq \|f\|_p \|g\|_q.$$

- For $p = q = 2$, this is Cauchy-Schwarz.

Proof idea of Holder

Proof idea: Young inequality

For $a, b \geq 0$ and conjugate exponents p, q ,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Apply this to

$$a = \frac{|f|}{\|f\|_p}, \quad b = \frac{|g|}{\|g\|_q},$$

then integrate.

Minkowski inequality

Theorem: Triangle inequality in L^p

For $1 \leq p < \infty$,

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

- This is the nontrivial part of the L^p norm proof.
- For $p = 1$, it follows directly from $|f + g| \leq |f| + |g|$.

Proof idea of Minkowski

Proof idea: Using Holder

For $p > 1$,

$$\|f + g\|_p^p = \int |f + g|^{p-1} |f + g|.$$

Use

$$|f + g| \leq |f| + |g|$$

and apply Holder to

$$|f| |f + g|^{p-1}, \quad |g| |f + g|^{p-1}.$$

Then divide by $\|f + g\|_p^{p-1}$ if nonzero.

L^p is a normed space

Theorem: Norm properties

For $1 \leq p \leq \infty$, $\|\cdot\|_p$ is a norm on equivalence classes of measurable functions:

- ① $\|f\|_p \geq 0$ and $\|f\|_p = 0$ iff $f = 0$ a.e.
- ② $\|af\|_p = |a| \|f\|_p$.
- ③ $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

Completeness of L^p

Theorem: Riesz-Fischer

For $1 \leq p \leq \infty$, $L^p(X, \mu)$ is complete. Hence it is a Banach space.

Proof idea: Proof idea for $p < \infty$

From a Cauchy sequence select a fast Cauchy subsequence. Show the telescopic series converges pointwise a.e. and in L^p . Then the whole sequence converges to the same limit.

Inclusions on finite measure spaces

Theorem: Finite measure inclusion

If $\mu(X) < \infty$ and $1 \leq q \leq p \leq \infty$, then

$$L^p(X) \subseteq L^q(X).$$

Moreover,

$$\|f\|_q \leq \mu(X)^{1/q-1/p} \|f\|_p.$$

Proof idea: Use Holder

Write $|f|^q \cdot 1$ and apply Holder.

Counterexample: inclusions fail on infinite measure spaces

Counterexample: L^p and L^q on $\mathbb{R}$

On $\mathbb{R}$, neither $L^p \subseteq L^q$ nor $L^q \subseteq L^p$ holds in general.

Example: Two directions

- $f(x) = \mathbf{1}_{(0,1)}x^{-a}$ can fail near zero.
- $g(x) = \mathbf{1}_{(1,\infty)}x^{-b}$ can fail at infinity.

Choose a, b so that membership differs for the desired p, q .

Convergence in L^p

Definition: L^p convergence

We say $f_n \rightarrow f$ in L^p if

$$\|f_n - f\|_p \rightarrow 0.$$

- This is stronger than convergence in measure on finite measure spaces.
- It does not imply pointwise convergence of the full sequence in general.
- It does imply the existence of a subsequence converging a.e.

A.e. convergence does not imply L^p convergence

Counterexample: Moving spikes

On $(0, 1)$ let

$$f_n = n^{1/p} \mathbf{1}_{(0, 1/n)}.$$

Then $f_n \rightarrow 0$ a.e., but

$$\|f_n\|_p = 1$$

for every n .

L^p convergence does not imply pointwise convergence

Counterexample: Typewriter sequence idea

There exists a sequence of indicators of intervals in $[0, 1]$ such that

$$\|f_n\|_p \rightarrow 0$$

for every finite p , but $f_n(x)$ does not converge for any x in $[0, 1]$.

Moral

L^p convergence is a norm convergence. It controls average size, not pointwise behavior at every index.

Cauchy in L^p

Definition: L^p Cauchy sequence

A sequence (f_n) is Cauchy in L^p if

$$\|f_n - f_m\|_p \rightarrow 0 \quad \text{as } n, m \rightarrow \infty.$$

Theorem: Completeness

Every Cauchy sequence in L^p has a limit in L^p .

Exercise: Use completeness

Show that if $\sum_n \|g_n\|_p < \infty$, then $\sum_n g_n$ converges in L^p .

L^2 as a Hilbert space

Definition: Inner product in L^2

For $f, g \in L^2(X, \mu)$ define

$$\langle f, g \rangle = \int_X fg \, d\mu$$

for real-valued functions.

Theorem: L^2 Hilbert space

$L^2(X, \mu)$ is a Hilbert space with norm

$$\|f\|_2 = \sqrt{\langle f, f \rangle}.$$

Orthogonality in L^2

Definition: Orthogonality

Functions $f, g \in L^2$ are orthogonal if

$$\int fg \, d\mu = 0.$$

Example: Simple orthogonality

On $[-1, 1]$,

$$\int_{-1}^1 x \, dx = 0.$$

Thus x is orthogonal to the constant function 1.

Projection onto constants

Example: Best constant approximation

On a finite measure space, find the constant c minimizing

$$\int_X (f - c)^2 d\mu.$$

The first-order condition is

$$\int_X (f - c) d\mu = 0.$$

Thus

$$c = \frac{1}{\mu(X)} \int_X f d\mu.$$

- This is projection onto the subspace of constant functions.

Projection onto a finite-dimensional span

Example: Normal equations

Let $V = \text{span}\{\varphi_1, \dots, \varphi_m\} \subset L^2$. The best approximation

$$g = \sum_{j=1}^m c_j \varphi_j$$

satisfies

$$\int (f - g) \varphi_i \, d\mu = 0, \quad i = 1, \dots, m.$$

Practical meaning

The error is orthogonal to every admissible direction.

Exam-style exercise: best constant

Exam-style exercise: Projection problem

Find the constant c minimizing

$$\int_0^1 (x^2 - c)^2 dx.$$

Solution

The best constant is the mean:

$$c = \int_0^1 x^2 dx = \frac{1}{3}.$$

Exam-style exercise: Holder

Exam-style exercise: Estimate an integral

Let $f \in L^3(0, 1)$ and $g \in L^{3/2}(0, 1)$. Prove that $fg \in L^1(0, 1)$ and estimate $\int |fg|$.

Solution

Since $1/3 + 2/3 = 1$, Holder gives

$$\int_0^1 |fg| \leq \|f\|_3 \|g\|_{3/2}.$$

Exam-style exercise: finite-measure inclusion

Exam-style exercise: $L^p \subseteq L^q$

Let $\mu(X) < \infty$ and $1 \leq q < p < \infty$. Prove that $L^p(X) \subseteq L^q(X)$.

Solution skeleton

Use Holder on

$$\int |f|^q = \int |f|^q \cdot 1$$

with exponents p/q and $p/(p - q)$.

Exam-style counterexample: inclusion fails

Exam-style exercise: Infinite measure

Give an example of a function in $L^2(1, \infty)$ but not in $L^1(1, \infty)$.

Solution

Take

$$f(x) = \frac{1}{x}.$$

Then

$$\int_1^\infty \frac{1}{x^2} dx < \infty, \quad \int_1^\infty \frac{1}{x} dx = \infty.$$

Approximation philosophy

Why approximation matters

Most functions are too complicated to handle directly. We approximate them by simpler functions, prove the result there, then pass to the limit.



choose the class suitable for the theorem

Simple approximation of nonnegative functions

Theorem: Approximation from below

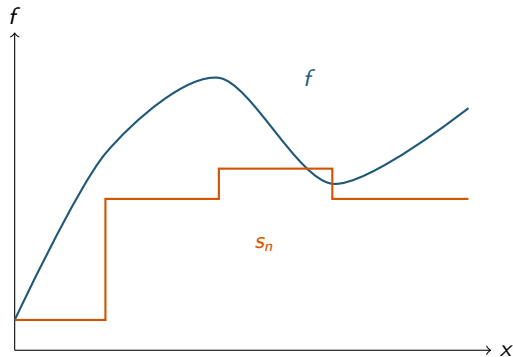
If $f \geq 0$ is measurable, there exists a sequence of simple functions s_n such that

$$0 \leq s_n \uparrow f.$$

Meaning

Every nonnegative measurable function can be built as the increasing limit of finite staircases.

Visual: staircase approximation



Intuition

Approximation becomes finer as the number of horizontal levels increases.

Density: practical meaning

Definition: Dense subset

A subset D of a normed space X is dense if every element of X can be approximated arbitrarily well by elements of D .

Meaning in L^p

A complicated function can often be approximated in L^p by simple, step, continuous or smooth functions.

Exam-style: simple functions are enough

Exam-style exercise: Conceptual proof

Explain why proving a positive linear identity first for indicators and then for simple functions is often enough to start a proof for nonnegative measurable functions.

Solution

Indicators generate simple functions by finite linear combinations. Nonnegative measurable functions are increasing limits of simple functions. The Monotone Convergence Theorem then allows the identity to pass from simple functions to general nonnegative measurable functions.

Common counterexample: missing domination

Counterexample: Prototype

The sequence

$$f_n = n\mathbf{1}_{(0,1/n)}$$

should be remembered as the standard counterexample for:

- pointwise convergence not implying convergence of integrals;
- convergence in measure not implying L^1 convergence;
- DCT failing when domination is absent.

Exam training: classify the theorem

Exam-style exercise: Choose and justify

For each sequence, state whether MCT, Fatou, DCT, or none is the best tool:

- ① $f_n = \mathbf{1}_{[0,n]}$ on $\mathbb{R}$.
- ② $g_n = \frac{\sin(nx)}{n}$ on $[0, 1]$.
- ③ $h_n = n\mathbf{1}_{(0,1/n)}$ on $[0, 1]$.

Solution

(1) MCT. (2) DCT with dominator 1. (3) Fatou gives an inequality; no equality theorem applies.

Exam training: finite measurable space

Exam-style exercise: Complete calculation

Let $X = \{1, 2, 3\}$, $\mu(\{1\}) = 2$, $\mu(\{2\}) = 1$, $\mu(\{3\}) = 3$, and $f(1) = 1$, $f(2) = -2$, $f(3) = 4$. Compute $\int f d\mu$ and $\|f\|_2$.

Solution

$$\int f d\mu = 1 \cdot 2 + (-2) \cdot 1 + 4 \cdot 3 = 12.$$

Also

$$\|f\|_2^2 = 1^2 \cdot 2 + (-2)^2 \cdot 1 + 4^2 \cdot 3 = 2 + 4 + 48 = 54,$$

so $\|f\|_2 = 3\sqrt{6}$.

Exam training: prove measurability and compute

Exam-style exercise: Mixed exercise

Let $A, B \in \mathcal{F}$ and define

$$f = 2\mathbf{1}_A - 3\mathbf{1}_B.$$

Show that f is measurable and compute $\int f d\mu$ when $\mu(A) = 4$ and $\mu(B) = 5$.

Solution

Indicators of measurable sets are measurable, and linear combinations of measurable functions are measurable. By linearity,

$$\int f d\mu = 2\mu(A) - 3\mu(B) = 8 - 15 = -7.$$

Exam training: find an a.e. representative

Exam-style exercise: Equivalence class

Let

$$f = \mathbf{1}_{\mathbb{Q} \cap [0,1]} + x \mathbf{1}_{[0,1] \setminus \mathbb{Q}}.$$

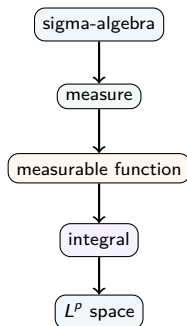
Find a simpler function equal to f almost everywhere and compute the integral.

Solution

Since $\mathbb{Q} \cap [0, 1]$ has measure zero, $f = x$ almost everywhere on $[0, 1]$. Therefore

$$\int_0^1 f \, d\lambda = \int_0^1 x \, dx = \frac{1}{2}.$$

End-of-module map



Final message

The module is one construction:

- ① choose measurable sets;
- ② measure them;
- ③ integrate measurable functions;
- ④ organize functions into normed spaces.

Module 2

Approximation, density and exam training

A recurring method

To prove a statement for general functions:

- ① prove it for indicators;
- ② extend it to simple functions by linearity;
- ③ extend it to nonnegative measurable functions by monotone approximation;
- ④ extend it to integrable functions by positive and negative parts.

Density of simple functions

Theorem: Simple functions are dense in L^p

For $1 \leq p < \infty$, every $f \in L^p$ can be approximated in L^p by simple functions.

Proof idea: Proof idea

Approximate $|f|$ by simple functions, truncate large values, and control the tail using integrability.

Truncation as a practical method

Example: Truncated functions

For $f \in L^p$, define

$$f^{(M)} = \max(-M, \min(f, M)).$$

Then

$$\|f^{(M)} - f\|_p \rightarrow 0$$

as $M \rightarrow \infty$.

Why

The error is supported on $\{|f| > M\}$ and is controlled by $|f|^p$.

Approximation by continuous functions: idea

On nice domains

On many standard spaces, integrable functions can be approximated by continuous or smooth functions in L^p .

- The exact theorem requires additional technical assumptions.
- For this course, the important idea is that irregular functions can be approximated in average size.
- Pointwise approximation and L^p approximation are different concepts.

Checklist for proof questions

Before writing the proof

Ask:

- ① Are the functions nonnegative?
- ② Is there monotonicity?
- ③ Is there domination by an integrable function?
- ④ Is the measure finite?
- ⑤ Are we working pointwise or modulo a.e. equality?
- ⑥ Which norm or topology is involved?

Common mistakes

Counterexample: Mistakes to avoid

- ① Confusing pointwise equality with a.e. equality.
- ② Applying DCT without finding a dominating function.
- ③ Swapping integrals without checking Tonelli or Fubini.
- ④ Assuming $L^p \subseteq L^q$ on infinite-measure spaces.
- ⑤ Treating a seminorm as a norm before quotienting by a.e. equality.

Exam exercise 1: sigma-algebra

Exam-style exercise: Generated sigma-algebra

Let $X = \{1, 2, 3, 4, 5\}$ and

$$A = \{1, 2, 3\}, \quad B = \{3, 4\}.$$

Find $\sigma(A, B)$.

Solution

Atoms are

$$A \cap B = \{3\}, \quad A \cap B^c = \{1, 2\},$$

$$A^c \cap B = \{4\}, \quad A^c \cap B^c = \{5\}.$$

The sigma-algebra consists of all unions of these four atoms.

Exam exercise 2: measurability

Exam-style exercise: Level-set criterion

Let f, g be measurable. Prove that

$$A = \{x : f(x) \leq g(x)\}$$

is measurable.

Solution

Since $f - g$ is measurable,

$$A = \{x : f(x) - g(x) \leq 0\}$$

is measurable by the half-line criterion.

Exam exercise 3: integral and null sets

Exam-style exercise: A.e. modification

Let $f \in L^1(X, \mu)$ and let $g = f$ a.e. Prove that $g \in L^1$ and

$$\int g \, d\mu = \int f \, d\mu.$$

Solution

Since $|g| = |f|$ a.e., $\int |g| = \int |f| < \infty$. Also $g - f = 0$ a.e., so $\int (g - f) = 0$.

Exam exercise 4: convergence theorem

Exam-style exercise: Limit of integrals

Compute

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{nx}{1 + n^2 x^2} dx.$$

Solution

Direct computation:

$$\int_0^1 \frac{nx}{1 + n^2 x^2} dx = \frac{1}{2n} \log(1 + n^2) \rightarrow 0.$$

DCT is not immediate from the naive bound because the peak moves with n .

Exam exercise 5: DCT with a correct bound

Exam-style exercise: Find domination

Show that

$$\lim_{n \rightarrow \infty} \int_0^{\infty} \frac{e^{-x}}{1+x/n} dx = 1.$$

Solution

Pointwise convergence gives $e^{-x}/(1+x/n) \rightarrow e^{-x}$. Also

$$0 \leq \frac{e^{-x}}{1+x/n} \leq e^{-x},$$

and e^{-x} is integrable on $(0, \infty)$. DCT gives the limit $\int_0^{\infty} e^{-x} dx = 1$.

Exam exercise 6: Fatou strictness

Exam-style exercise: Construct strict Fatou

Find nonnegative measurable f_n such that $f_n \rightarrow 0$ pointwise but

$$\liminf_n \int f_n d\mu > 0.$$

Solution

On $\mathbb{R}$, take $f_n = \mathbf{1}_{[n, n+1]}$. Then $f_n \rightarrow 0$ pointwise and $\int f_n = 1$.

Exam exercise 7: L^p norm

Exam-style exercise: Compute a norm

For $f(x) = x$ on $[0, 1]$, compute $\|f\|_p$ for $1 \leq p < \infty$.

Solution

$$\|f\|_p = \left(\int_0^1 x^p dx \right)^{1/p} = \left(\frac{1}{p+1} \right)^{1/p}.$$

Exam exercise 8: L^2 projection

Exam-style exercise: Projection onto linear functions

Find a, b minimizing

$$\int_0^1 (x^2 - a - bx)^2 dx.$$

Solution skeleton

The error is orthogonal to 1 and x :

$$\int_0^1 (x^2 - a - bx) dx = 0,$$

$$\int_0^1 x(x^2 - a - bx) dx = 0.$$

Solve the resulting 2×2 linear system.

Exam exercise 8: solution

Solving the system

The equations are

$$\frac{1}{3} - a - \frac{b}{2} = 0, \quad \frac{1}{4} - \frac{a}{2} - \frac{b}{3} = 0.$$

Solving gives

$$b = 1, \quad a = -\frac{1}{6}.$$

Thus the best linear approximation is

$$-\frac{1}{6} + x.$$

Exam exercise 9: counterexample to inclusion

Exam-style exercise: Infinite measure

For $1 < p < q$, construct a function on $(1, \infty)$ belonging to L^q but not to L^p .

Solution

Take $f(x) = x^{-a}$ with

$$\frac{1}{q} < a \leq \frac{1}{p}.$$

Then $aq > 1$ but $ap \leq 1$.

Exam exercise 10: norm zero

Exam-style exercise: Equivalence classes

Let $f \in L^p$ and suppose $\|f\|_p = 0$. Prove that $f = 0$ a.e.

Solution

Since $|f|^p \geq 0$ and

$$\int |f|^p = 0,$$

the theorem for nonnegative functions implies $|f|^p = 0$ a.e., hence $f = 0$ a.e.

Exam exercise 11: convergence in L^p

Exam-style exercise: A convergence test

Let $f_n = x^n$ on $[0, 1]$. Does $f_n \rightarrow 0$ in L^p for finite p ?

Solution

Yes:

$$\|x^n\|_p^p = \int_0^1 x^{np} dx = \frac{1}{np+1} \rightarrow 0.$$

Thus $\|x^n\|_p \rightarrow 0$.

Exam exercise 12: pointwise versus L^p

Exam-style exercise: Moving spikes

Let $f_n = n\mathbf{1}_{(0,1/n)}$ on $(0,1)$. For which $p \geq 1$ does $f_n \rightarrow 0$ in L^p ?

Solution

$$\|f_n\|_p^p = \int_0^{1/n} n^p dx = n^{p-1}.$$

For $p \geq 1$, this does not go to 0. Hence there is no L^p convergence to zero.

Summary of Module 2

Core concepts

- Sigma-algebras specify which sets are measurable.
- Measures assign countably additive sizes to measurable sets.
- Measurable functions are compatible with measurable sets.
- Lebesgue integration is built from simple functions.
- MCT, Fatou and DCT are the main limit theorems.
- L^p spaces are Banach spaces; L^2 is a Hilbert space.

What to review before the next module

- ① Definitions of sigma-algebra, measure, measurable function.
- ② Construction and properties of the Lebesgue integral.
- ③ MCT, Fatou and DCT, including counterexamples.
- ④ L^p membership tests for power functions.
- ⑤ Holder and Minkowski inequalities.
- ⑥ A.e. equality and equivalence classes.
- ⑦ Orthogonality and projection in L^2 .

Recommended study method

For each theorem, memorize the hypotheses by building a counterexample where one hypothesis fails.

End of Module 2

Measure, Integration and L^p Spaces

Next: probabilistic structures and random variables